Local Income Taxes and Residential Sorting:

Evidence from LEHD Origin-Destination Data

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Abstract

Do local income tax increases cause residential sorting across municipal boundaries? Using LEHD Origin-Destination data linking each worker's residence to their workplace at municipal resolution, I estimate the effect of earned income tax rate changes on the locally-resident share of a city's workforce in Ohio and Pennsylvania, 2002–2019. Rate increases reduce this share by 0.5–0.8 percentage points, building gradually over 7–8 years. The response is asymmetric: rate increases push residents out, but rate decreases do not bring them back. Pennsylvania's Act 32 (2012), which consolidated EIT collection and mandated employer withholding, provides a natural experiment: the sorting response is a precise zero before 2012 and highly significant after, confirming that workers only respond to taxes they actually pay. Tract-level decomposition reveals that the sorting operates through cross-boundary relocation rather than within-city reshuffling, that school district EIT rates attract residents (fully offsetting municipal rate repulsion in high-SD areas), and that the boundary gradient is null—sorting is city-wide, not concentrated at municipal edges. Despite the behavioral response, rate increases are fiscally productive: EIT revenue rises 67% per percentage point, total tax revenue rises 30%, and population and property values are unaffected. The residential sorting induced by local income taxes is real but too small to erode the tax base.

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1 Introduction

The Tiebout hypothesis—that households "vote with their feet" by sorting across jurisdictions offering different bundles of taxes and public services (Tiebout, 1956)—remains a central organizing principle in local public finance. A large empirical literature documents that state-level tax differences influence the residential choices of high earners (Kleven et al., 2020; Young et al., 2016) and that local sales taxes generate cross-boundary shopping responses (Agrawal, 2015). Yet surprisingly little evidence exists on whether *local* income tax differences cause residential sorting *within* metropolitan areas, where moving costs are low and tax boundaries are crossed by a short relocation.

This gap persists for two reasons. First, most jurisdictions that levy local income taxes (municipalities in Ohio, Pennsylvania, and a handful of other states) are too small to appear in standard migration datasets. The IRS Statistics of Income tracks county-to-county migration, but local income taxes vary at the sub-county level; the American Community Survey measures place-level population but does not link workers to their employers. Second, local income tax systems feature credit mechanisms, overlapping school district taxes, and varying enforcement regimes that complicate the mapping from statutory rates to effective tax burden. A naive comparison of tax rates and population changes conflates tax-motivated sorting with amenity valuation, fiscal capacity differences, and enforcement quality.

This paper addresses both challenges using LEHD Origin-Destination Employment Statistics (LODES), the only administrative data source that links each worker's residence tract to their workplace tract at annual frequency. From the approximately 24 million tract-pair-year observations per state, I construct a municipal-level outcome—the share of a city's workers who also live there—and a set of disaggregated tract and city-pair panels that decompose the sorting response into distinct adjustment channels. The two-state design exploits complementary sources of variation: Ohio provides a clean extensive margin (taxed municipalities versus untaxed townships), while Pennsylvania's universal EIT with rate variation, overlapping school district taxes, and the natural experiment of Act 32 (2012) enforcement

consolidation enables tests of mechanism and heterogeneity.

The main finding is that local income tax rate increases reduce the locally-resident share of a city's workforce by 0.5–0.8 percentage points, building gradually over 7–8 years. The response is asymmetric—rate increases push residents out, but rate decreases do not bring them back—and is entirely concentrated after 2012, when Pennsylvania's Act 32 made employer withholding mandatory and the tax became binding. These results are robust across estimators (Sun-Abraham, Sun and Abraham, 2021), continuous treatment models (de Chaisemartin et al., 2024), and both states.

Decomposing the aggregate response reveals three findings that depart from the canonical Tiebout prediction. First, the sorting operates through cross-boundary relocation (workers leave the city entirely) rather than within-city reshuffling, and the boundary gradient is null—the effect is city-wide, not concentrated at municipal edges. Second, school district EIT rates have the *opposite* effect: higher school district rates attract residents, completely offsetting municipal rate repulsion in high-SD areas. Act 1 of 2006 authorized SD rate increases specifically for property tax relief, though our data cannot confirm that property taxes actually fell (the PDE fiscal data postdates most SD rate changes). This explains why studies using total EIT rates find null results. Third, commuter destination choices are unrelated to the destination's municipal EIT rate, exactly as the credit system predicts: workers who keep their job gain no tax savings from moving.

Despite the real behavioral response, rate increases are fiscally productive. EIT revenue rises 67% per percentage point of rate increase, total tax revenue rises 30%, and population and property values are unaffected. The residential sorting induced by local income taxes is too small to erode the tax base. For municipal policymakers considering a rate increase, the message is clear: you will lose some residents at the margin, but you will keep the revenue.

2 Institutional Background

Local income taxes are one of the few fiscal instruments available to municipalities in Ohio and Pennsylvania. Unlike property taxes, which are paid by property owners and capitalized into home values, local income taxes fall directly on workers and vary across municipal boundaries within the same metropolitan area. This creates sharp spatial discontinuities in tax burden that can influence where people choose to live and work.

2.1 Ohio Municipal Income Tax

Ohio authorizes municipalities (cities and villages) to levy a municipal income tax on earned income under Ohio Revised Code Chapter 718. Unincorporated townships cannot levy the tax, creating a clean extensive-margin contrast between taxed and untaxed jurisdictions within the same county. As of 2019, approximately 685 municipalities levy the tax at rates ranging from 0.5% to 3.0%, with 1.0% and 1.5% being the most common. Our panel (2002–2019) contains 302 rate increases and 45 rate decreases.

Ohio operates a credit system: residents who work in a municipality with a higher tax rate than their residence receive a credit against their home municipality's tax for taxes paid to the work municipality. However, this credit system has historically been administered at the municipal level without centralized enforcement, and the credit is often partial (capped at the home municipality's rate or a fraction thereof). This means that for Ohio workers, the interaction between home and work municipality rates determines the effective tax burden, but imperfect crediting can leave workers paying more than the statutory maximum of either rate alone.

2.2 Pennsylvania Earned Income Tax

Pennsylvania's Earned Income Tax (EIT) is mandatory for all 2,500+ municipalities under the Local Tax Enabling Act (Act 511 of 1965). The EIT applies to all earned income with

no minimum threshold; retirees are exempt. The standard cap is 1%, typically split 0.5% municipal and 0.5% school district. Home Rule municipalities may exceed this cap, with Philadelphia operating under a separate Sterling Act authority at approximately 3.8%. The EIT is the primary revenue lever for most Pennsylvania municipalities: property tax millage caps are binding in many jurisdictions, and EIT rate changes are uncorrelated with property tax changes ($r = -0.04$). Our panel contains 291 rate increases and 162 rate decreases.

The Credit System. A critical institutional feature is the credit mechanism governing how EIT is allocated between a worker's municipality of residence and municipality of employment. A worker's total EIT liability equals the *higher* of the resident rate and the work (nonresident) rate:

$$\tau_w^{\text{total}} = \max\left(\tau_w^{\text{res}},\ \tau_w^{\text{work}}\right) \tag{1}$$

If a worker lives and works in the same municipality, they simply pay that municipality's rate. If they live in a low-rate municipality but work in a high-rate one, the employer withholds the work rate; the home municipality receives nothing beyond its own rate (or the residual after the work municipality's claim). Crucially, if a worker moves from a high-rate municipality to a lower-rate one while keeping the same job, their *total tax liability does not change*—they continue to pay the work municipality's rate through employer withholding, with the credit system ensuring the work municipality collects the difference.

This has a direct implication for interpreting residential sorting. Under the credit system, a worker who changes residence but not workplace gains no tax savings from the move. The only way to reduce one's EIT burden is to change both residence and workplace to a lower-rate jurisdiction, or to move one's residence to a municipality whose rate equals or exceeds the work rate (so the residence rate governs and the work municipality receives nothing).

Act 32: Tax Collection Consolidation. Before 2012, EIT collection was radically decentralized: each of Pennsylvania's 2,500+ municipalities appointed its own tax officer. Non-

resident EIT enforcement was weak because no centralized system linked employers, workers, and municipalities. A worker who moved out of a rate-raising municipality could, in practice, avoid paying the nonresident portion to the work municipality even though they legally owed it. The credit system described in Equation 1 existed on paper but was not effectively enforced.

Act 32 of 2008 (effective January 1, 2012) transformed this system by consolidating collection into approximately 67 county-based Tax Collection Districts (TCDs), each with a single appointed collector. The law mandated standardized employer withholding at the higher of the employee's resident and work rate. This had two consequences. First, it made the EIT *binding*: rate changes now mechanically appear in every worker's paycheck through employer withholding, regardless of whether they reside in the taxing municipality. Second, it eliminated the possibility of de facto tax avoidance through residential relocation. Before Act 32, moving away from a rate-raising city could effectively reduce a worker's tax burden because the work municipality lacked the enforcement capacity to collect from nonresidents. After Act 32, the credit system is fully enforced, and moving away while keeping the same job provides no tax relief.

School District EIT. School districts in Pennsylvania also levy an EIT under Act 511, typically at 0.5% (the other half of the standard 1% cap). Act 1 of 2006 (the Taxpayer Relief Act) additionally authorized school districts to exceed the 0.5% cap specifically to fund property tax relief. This generated a wave of school district rate increases in 2006–2011, creating rate variation that is largely independent of municipal fiscal decisions: the correlation between municipal and school district EIT rates across our panel is -0.12. Since every census tract sits in both a municipality and a school district, workers face two overlapping EIT rates that can change independently, providing a natural decomposition of the residential sorting response into municipal and school district components.

3 Conceptual Framework

We develop a framework for how local income tax changes affect the spatial distribution of workers across municipalities. The key institutional features described above—the credit system, Act 32 enforcement, and overlapping municipal and school district taxes—generate distinct predictions about *who* moves, *where* they go, and *why*.

3.1 Three Channels of Adjustment

Consider a municipality A that raises its EIT rate. Workers currently employed in A can respond through three channels, each with different implications for their tax burden:

Channel 1: Residential relocation without job change. A worker moves their residence from A to another municipality B but continues working in A. Under the credit system (Equation 1), this worker's total EIT liability is $\max(\tau_B, \tau_A)$. If $\tau_A > \tau_B$—the typical case when A just raised its rate—the worker still pays τ_A. *Moving provides zero tax savings.* This channel reduces A's share_live_in (our primary outcome) but does not reduce the worker's tax burden.

Channel 2: Complete relocation (residence and job). A worker moves both residence and job from A to municipality B, where $\tau_B < \tau_A$. Their new liability is τ_B, yielding genuine tax savings. This is the canonical Tiebout sorting response. This channel reduces both A's resident worker count and total employment. In our data, it appears as a decline in the $(w = A, h = A)$ cell of the origin-destination matrix and an increase in the $(w = B, h = B)$ cell for nearby municipalities B.

Channel 3: Compositional change in new hires. When A creates new positions or replaces departing workers, the new hires may choose not to live in A given the higher rate. They take the job in A but live in B. Under the credit system, they pay $\max(\tau_B, \tau_A) = \tau_A$,

so this choice reflects non-tax considerations (housing costs, school quality, neighborhood preferences). This channel reduces A's share_live_in even though no existing resident moved.

3.2 Predictions

These channels generate testable predictions that differ from a naive Tiebout model:

Prediction 1: Enforcement matters. If the sorting response is driven by workers optimizing their tax burden (Channel 2), the effect should depend on whether the tax is actually collected. Before Act 32, when nonresident enforcement was weak, a rate increase was partly a paper change. After Act 32, it mechanically reduces take-home pay. We therefore expect the sorting response to be *stronger* after 2012, not because moving became more attractive (it became less so under full credit enforcement), but because the rate increase itself became binding. Workers only respond to taxes they actually pay.

Prediction 2: Destination rates should not matter for stayers. Workers who change residence but keep the same job (Channel 1) gain no tax advantage from choosing a lower-rate destination. If the rate gap between work and home municipality predicts commuter flows, it reveals a non-tax channel: housing cost capitalization, amenity perception, or fiscal quality signaling. If destination rates are irrelevant to commuter flows, the credit system is operating as designed and workers are not confused about their tax liability.

Prediction 3: Complete relocators should choose lower-rate destinations. Workers who change both residence and job (Channel 2) *can* reduce their tax burden by choosing a lower-rate municipality. If cities near a rate-raising municipality experience gains in resident workers, and those gains are larger for lower-rate destinations, this confirms tax-motivated complete relocation.

Prediction 4: School district taxes should have opposite effects. Municipal EIT rate increases are typically associated with fiscal stress—the municipality needs revenue. School district EIT rate increases, particularly those authorized under Act 1 of 2006, were designed to fund property tax relief and school spending. Higher school district EIT rates may therefore signal a *valued amenity* (better schools, potentially lower property taxes) rather than fiscal extraction. If so, school district rate increases should *attract* residents, offsetting the repulsive effect of municipal rate increases. This decomposition is possible because municipal and school district rates vary independently ($r = -0.12$) and every tract sits in both jurisdictions simultaneously.

Prediction 5: Sorting should occur at municipal boundaries. If EIT rate changes cause cross-boundary residential sorting, the effect should be concentrated at the edges of municipalities where moving a short distance crosses a tax boundary. Interior tracts, far from any municipal boundary, should show weaker responses. The absence of a boundary gradient would suggest that the "sorting" is actually a city-wide population decline unrelated to spatial tax arbitrage.

4 Data

4.1 Data Construction Pipeline

The analysis requires linking three administrative systems that do not share common identifiers: LEHD/LODES employment records (Census tract geography), Ohio municipal tax rates (Census place geography), and Pennsylvania EIT rates (DCED municipality IDs). We construct the analysis panels through the following steps:

1. **Scrape PA DCED MunStats tax rates.** PA does not publish a downloadable panel of historical EIT rates. The only source is the DCED MunStats portal, which serves one XLS file ($\sim$12 MB) per year via an ASP.NET form. A Python script submits

requests for each year 2000–2026, extracting resident and nonresident EIT rates, school district EIT rates, and Local Services Tax for all ∼2,500 municipalities.

2. **Scrape PA DCED Annual Financial Reports.** Fiscal controls require municipality-level revenue and expenditure data. The DCED Annual Financial Reports (same portal, different report) provide 71 columns per municipality-year including tax revenue by type, police and streets expenditure, debt, population, and market value of real estate. Years 2002–2023.

3. **Download Ohio rate table.** Unlike PA, Ohio publishes a complete rate history as a single CSV ("The Finder") with start date, end date, municipality FIPS, and rate for every municipal income tax spell. Direct download.

4. **Download LODES origin-destination files.** LODES provides the only administrative data linking each worker's residence tract to their workplace tract at annual frequency. We download main files (intra-state flows) and auxiliary files (cross-state flows where the workplace is in-state but the residence is out-of-state) for OH and PA, 2002–2021. The auxiliary files are necessary because excluding cross-state commuters would mechanically inflate share_live_in for border municipalities. Segments include all jobs (S000), earnings categories (SE01, SE03), age groups (SA01–SA03), and federal employment (JT04, used as a placebo). Reference date: April 1 of each year.

5. **Expand Ohio rates to municipality × year.** The Finder reports rate spells (start/end dates), not a panel. We expand to municipality × year, where a rate is active in year Y if it was in effect on April 1 of year Y—aligned to the LODES reference date so that the rate a worker faces and the OD flow that records their location refer to the same point in time. Compute year-over-year rate changes and flag each municipality's first increase year (the treatment cohort for Sun-Abraham).

6. **Crosswalk PA DCED to Census geography.** LODES uses Census tract geogra-

phy; PA rates use DCED municipality IDs with no Census equivalent. PA municipalities are county subdivisions (COUSUBFP), not Census places, so a standard place-level crosswalk does not work. We match DCED municipality names to TIGER/Line COUSUBFP codes using normalized name + county. Match rate: 99.6%. The 9 unmatched municipalities are boroughs and townships that were merged or dissolved before the 2020 Census.

7. **Build tract-to-municipality crosswalks.** LODES reports flows at the tract level, but treatment varies at the municipality level. We assign each tract to a municipality by spatially joining tract centroids to 2020 TIGER/Line boundaries (OH places, PA county subdivisions). This is necessary for aggregating tract-level OD flows to the municipality panel and for the tract-level decomposition in Stage 2.

8. **Compute city-to-city neighbor pairs.** The relocation and commuter pair models require defining which municipalities are "neighbors." We compute centroid-to-centroid distances between all municipality pairs and retain those within 80 km. This radius captures the plausible commuting and relocation zone while excluding distant municipalities whose labor markets do not overlap.

9. **Aggregate LODES to municipality-level OD panels.** The raw LODES data has ∼24 million tract-pair-year observations per state. We aggregate to work municipality × year, counting total jobs, live-and-work (workers who also reside in the work municipality), commuters from other in-state municipalities, and cross-state commuters. This produces share_live_in, the primary outcome: the fraction of a city's workforce that also lives there.

10. **Merge rates into OD panels.** The final step joins rates to OD panels, producing the analysis-ready municipality × year datasets. OH municipalities with no rate history are assigned rate = 0, forming the no-tax control group. PA municipalities without a

DCED-to-Census match are dropped (9 of 2,505 municipalities, all merged or dissolved before 2020).

4.2 Data Sources

LODES Origin-Destination Data. LEHD LODES Version 8, 2002–2019. Tract-level residence–workplace links derived from state Unemployment Insurance wage records and the Quarterly Census of Employment and Wages. We trim 2020–2021 due to COVID-related disruptions in both employment patterns and LODES coverage. Noise infusion is applied at the census block level; aggregation to the municipality level substantially reduces measurement error.

Municipal Tax Rate Panels. Ohio: Department of Taxation "The Finder" CSV. Rate spells with start/end dates for all 685 taxing municipalities. Direct PLACEFP join to LODES. Pennsylvania: DCED MunStats, scraped annually. Resident and nonresident municipal EIT rates, school district EIT rates for all ∼2,500 municipalities. Name-based crosswalk to Census COUSUBFP (99.6% match rate).

Municipal Financial Reports. PA DCED Annual Financial Reports, 2002–2019 (trimmed from 2002–2023 to exclude COVID/ARPA fiscal distortion). 71 columns per municipality-year including EIT revenue, real estate tax revenue, police expenditure, streets expenditure, total debt, state intergovernmental aid, population, and assessed market value of real estate.

Crime. FBI Uniform Crime Reports, Offenses Known and Clearances by Arrest (Jacob Kaplan concatenated files), 2002–2019. Monthly agency-level counts of Part I index crimes (murder, robbery, aggravated assault, burglary, larceny-theft, motor vehicle theft). Aggregated to annual totals per agency. Matched to PA municipalities via `fips_place_code` (93.5% agency match rate).

Building Permits. Census Bureau Building Permits Survey (BPS), place-level annual files, 2002–2019. Total housing units permitted by structure type (1-unit, 2-unit, 3–4 unit, 5+ unit). PA municipalities matched via `fips_mcd_code` = COUSUBFP (93.3% MCD match rate).

4.3 Sample Restrictions and Transformations

1. **Sample period: 2002–2019.** LODES coverage begins in 2002. We trim at 2019 because COVID disrupted both commuting patterns (LODES 2020–2021) and municipal finances (ARPA transfers distort AFR data 2020–2023). Including these years would conflate pandemic-driven relocation with tax-motivated sorting.

2. **Minimum employment threshold: 50 total jobs.** Small municipalities have share_live_in values dominated by sampling noise (a single worker moving changes the ratio by $1/N$). The 50-job threshold balances precision against sample size. Sensitivity analysis (Section 7) confirms stability across thresholds from 10 to 500.

3. **Rate outlier filters.** PA DCED reports the Local Services Tax (a flat dollar amount) in the same field as the EIT rate (a percentage) in some years. Dropping rates $>$ 5% removes these data entry errors without affecting any real EIT rate (the highest standard EIT is $\sim$1%; only Philadelphia exceeds 3%).

4. **April 1 alignment.** LODES records employment as of April 1. A rate change that takes effect mid-year could be misattributed if not aligned. We assign each municipality-year the rate in effect on April 1, ensuring the tax rate a worker faces and the OD flow recording their location refer to the same date.

5. **Outcome variables.** The primary outcome is $\text{share_live_in}_{it} = \text{live_and_work}_{it}/\text{total_jobs}_{it}$, the locally-resident fraction of a city's workforce. Unlike population counts (which mix retirees, students, and non-workers) or IRS migration data (which is county-level), share_live_in isolates the residential choices of active workers at municipal resolution.

To decompose the share response into levels, we also estimate effects on log(resident workers), log(nonresident commuters), and log(total jobs) separately. Stage 2 introduces additional outcomes: log(commuter flow) for city-pair panels, log(resident workers) for the neighbor exposure model, and the within-city residence share at the tract level. Fiscal outcomes include log EIT revenue, log total tax revenue, log total revenue, log police expenditure, log population, and log market value of real estate from the PA DCED Annual Financial Reports. Pre-treatment balance tests cover eleven observables including crime and building permits.

6. **Treatment definitions.** We use three treatment measures to test different margins. *Binary*: 1 after a municipality's first rate increase, identifying the extensive-margin sorting response and enabling Sun-Abraham estimation. *Continuous*: the current rate level, capturing the intensive margin via DCDH and TWFE. *Cumulative dose*: current rate minus baseline, isolating the net rate change from the rate level for treated-only dose-response models.

7. **Control groups.** OH offers two natural control groups: (a) municipalities that never adopted a tax (clean extensive-margin contrast, 104 cities) and (b) municipalities with a stable rate (intensive-margin contrast, 403 cities). PA has only (b) because EIT is universal (2,261 never-changed municipalities). The no-tax control in OH provides the cleanest identification; the stable-rate control in PA provides the largest donor pool for Sun-Abraham.

8. **Fiscal controls.** Rate increases may coincide with other fiscal changes (police hiring, debt issuance) that independently affect residential attractiveness. We include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid as time-varying controls from the PA DCED AFR. The main estimates are unchanged with and without these controls (Table 19).

9. **Neighbor exposure.** To test whether workers displaced from rate-raising cities relo-

cate to nearby lower-rate municipalities (Channel 2), we compute each municipality's average neighbor EIT rate within 80 km. A positive coefficient on neighbor rates indicates that cities absorb workers when surrounding municipalities raise rates.

10. **Commuter pair panel.** The municipal-level outcome cannot distinguish residential relocation from complete job relocation. We disaggregate LODES to (work city A, home city B, year) pairs within 80 km, enabling direct tests of how rate changes in the work municipality affect the flow of commuters from specific home municipalities. Pair fixed effects absorb time-invariant distance, amenity differentials, and initial rate gaps.

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
Panel A: Pennsylvania (2002–2019)					
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Panel B: Ohio (2002–2019)					
share_live_in	0.119	0.099	0.085		
Municipal tax rate (%)	1.222	1.500	0.821	0	3.000
Panel dimensions					
PA municipality × years	26,650 (1,479 municipalities)				
OH municipality × years	7,740 (430 municipalities)				

Notes: Municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. OH municipalities are Census places; PA municipalities are county subdivisions.

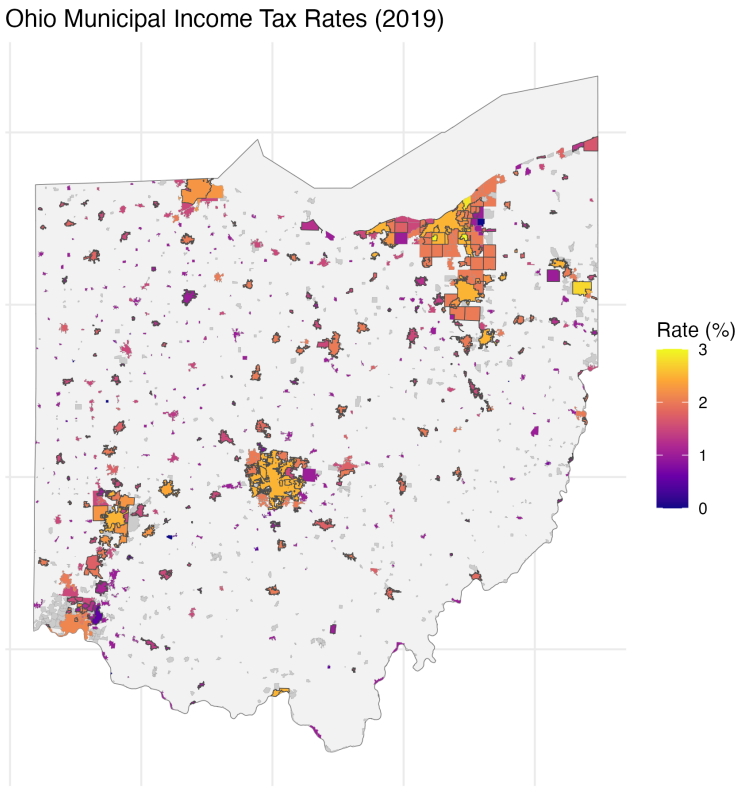


Figure 1: Ohio Municipal Income Tax Rates (2019). Bold outline = municipalities that raised rates during the sample period. Grey fill = municipalities with no municipal income tax.

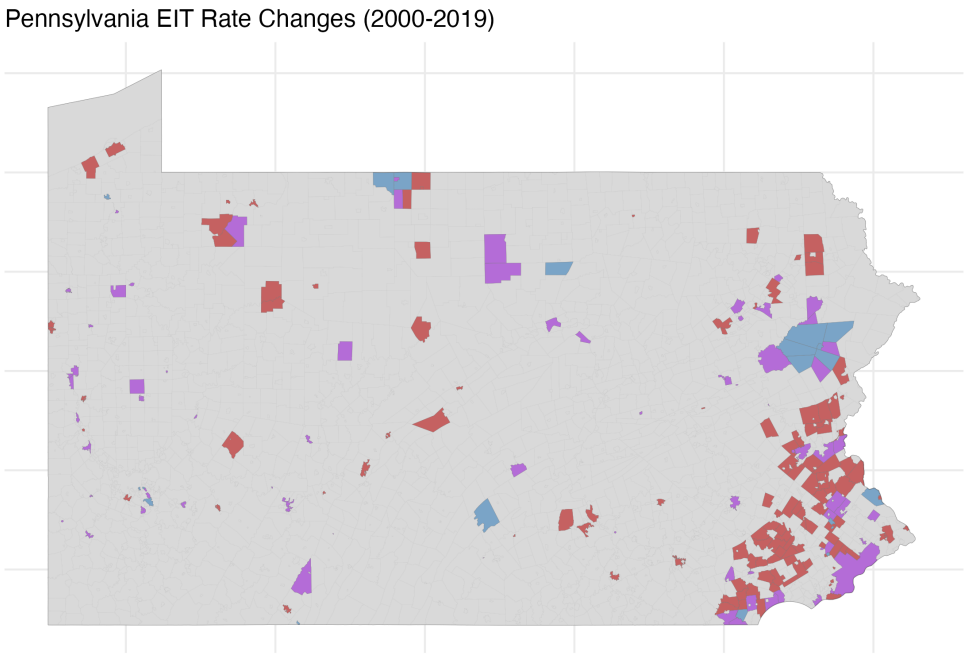


Figure 2: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

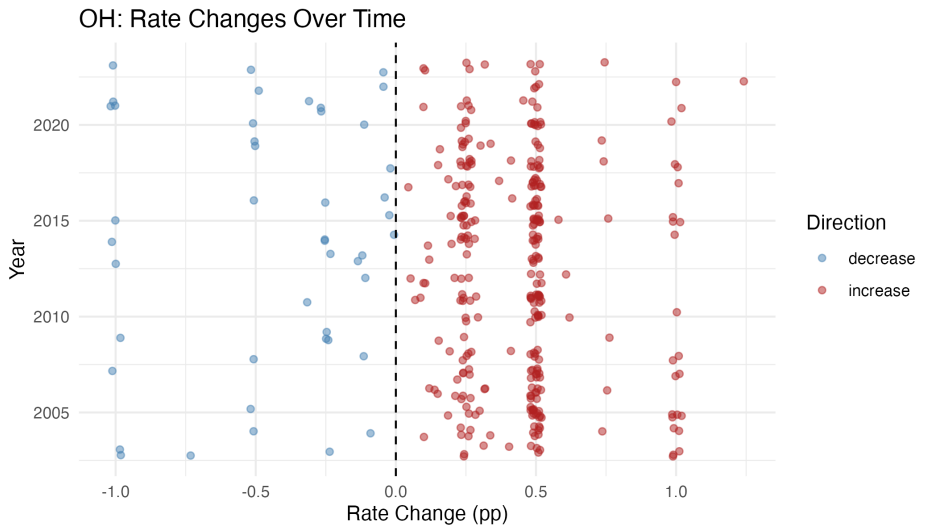


Figure 3: Ohio: Distribution of Municipal Income Tax Rate Changes Over Time

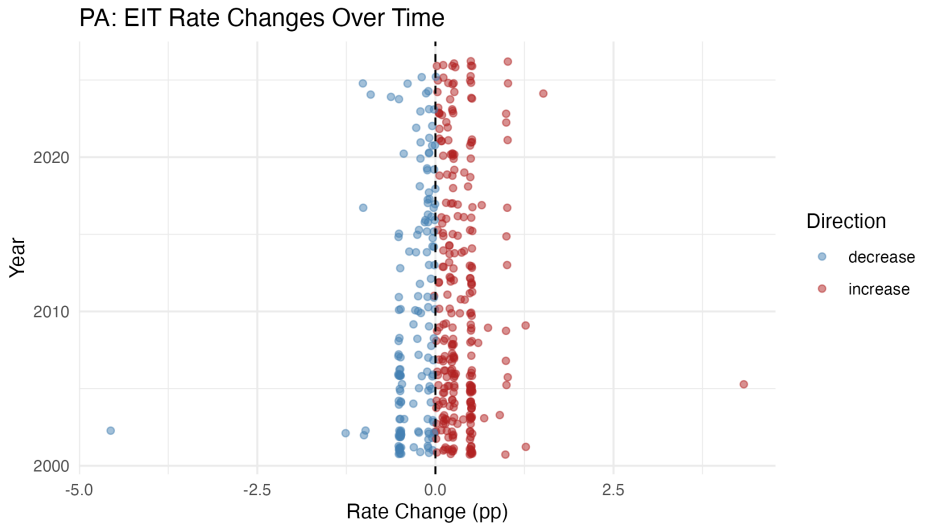


Figure 4: Pennsylvania: Distribution of EIT Rate Changes Over Time

5 Empirical Strategy

We organize the analysis in two stages. The first stage establishes the baseline sorting response using municipal-level outcomes, building from the extensive margin (tax adoption) through the intensive margin (rate changes) to asymmetry. The second stage uses tract-level origin-destination data to decompose the sorting response into the three channels described in Section 3.1 and to test the predictions from Section 3.2.

All specifications include municipality (or tract, or pair) and year fixed effects, with standard errors clustered at the municipality level. The municipality fixed effects absorb time-invariant differences across jurisdictions; the year fixed effects absorb aggregate shocks common to all municipalities (e.g., business cycles, statewide policy changes). Identification comes from within-municipality rate variation over time.

5.1 Stage 1: Municipal-Level Sorting

For binary treatments under staggered timing, we use the Sun and Abraham (2021) interaction-weighted estimator:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \cdot D_{it}^{\ell} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{2}$$

where Y_{it} is share_live_in for municipality i in year t, D_{it}^{ℓ} indicates municipality i being ℓ years from its first treatment, and $\mathbf{X}_{it}$ is a vector of fiscal controls (police expenditures, streets, debt service, and state aid; available for PA). Pre-treatment coefficients ($\ell < -1$) test parallel trends. For continuous treatments (the actual tax rate), we use the de Chaisemartin et al. (2024) estimator, which handles both staggered timing and varying treatment intensity.

Q1: Does adopting a local income tax cause residential sorting? Ohio municipalities can adopt a local income tax; unincorporated townships cannot. We compare municipalities that adopted a tax during the panel (going from 0 to a positive rate) against no-tax townships using Equation 2. This is the cleanest test of the extensive margin: a discrete switch from no tax to having one. Only Ohio provides this variation; all Pennsylvania municipalities levy an EIT throughout the panel.

Q2: Do rate increases cause additional sorting? Conditional on already having a tax, does raising the rate push additional residents out? We estimate Equation 2 with rate increasers as treated. For Ohio, we use no-tax townships as controls (stable-rate taxed cities produce null results, likely because they share unobserved urban characteristics with treated cities). For Pennsylvania, controls are the 2,209 municipalities that never changed their EIT rate. We also estimate the DCDH continuous treatment model to test whether rate *magnitude* matters beyond binary treatment status.

Q3: Are rate changes asymmetric? We test whether increases and decreases have symmetric effects using a split cumulative dose specification:

$$Y_{it} = \alpha_i + \gamma_t + \beta^+ \cdot R_{it}^+ + \beta^- \cdot R_{it}^- + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{3}$$

where R_{it}^+ and R_{it}^- are cumulative rate increases and decreases from baseline. If $|\beta^+| > |\beta^-|$, residential location decisions are sticky: rate increases push residents out, but rate decreases do not pull them back. We also estimate Equation 2 separately for rate decreasers versus never-changed controls.

Act 32 structural break. Prediction 1 implies that the sorting response should differ before and after the 2012 consolidation of tax collection. We test this by interacting the treatment variable with pre- and post-Act 32 indicators:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}} \cdot \tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}} \cdot \tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it} \tag{4}$$

Under Prediction 1, $\beta^{\text{pre}} \approx 0$ (no response when the tax is not effectively collected) and $\beta^{\text{post}} < 0$ (sorting begins when the tax becomes binding through employer withholding).

5.2 Stage 2: Tract-Level Decomposition

The municipal-level outcome share_live_in captures the net effect of all three adjustment channels but cannot distinguish between them. To decompose the response, we exploit the tract-level structure of LODES origin-destination data, which records job counts for every $(w_{\text{tract}}, h_{\text{tract}}, \text{year})$ triplet. We aggregate these flows to the city-pair level—(A, B, t) for work municipality A and home municipality B—and construct three analytic panels.

Q4: Do workers leave entirely, or just change where they sleep? *(Tests Predictions 2 and 3: destination choice and complete relocation.)*

For each municipality B, we construct a *resident worker panel* counting workers who both live and work in B. We then measure B's "exposure" to neighboring rate increases:

$$\ln(\text{ResWorkers}_{Bt}) = \alpha_B + \gamma_t + \beta_1 \cdot \tau_{Bt} + \beta_2 \cdot \bar{\tau}_{Bt}^{\text{nbr}} + \varepsilon_{Bt} \tag{5}$$

where τ_{Bt} is B's own EIT rate and $\bar{\tau}_{Bt}^{\text{nbr}}$ is the average rate among B's neighbors (within 80 km). If $\beta_1 < 0$, higher own rates reduce resident workers (consistent with Channels 1–3). If $\beta_2 > 0$, cities absorb resident workers when neighbors raise rates—evidence of Channel 2 (complete relocation toward the lower-rate city).

To test Prediction 2 (destination rates irrelevant for stayers), we estimate a *commuter pair model*:

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_1 \cdot \tau_{At}^{w} + \beta_2 \cdot \tau_{Bt}^{h} + \varepsilon_{ABt} \tag{6}$$

for all commuter pairs ($A \neq B$) within 80 km, where Flow_{ABt} is the number of workers employed in A and living in B. Under the credit system, workers who keep their job in A pay $\max(\tau_A, \tau_B) = \tau_A$ regardless of where they live (when $\tau_A > \tau_B$). If Prediction 2 holds, $\beta_2 \approx 0$: the home municipality's rate does not predict commuter flows, because it provides no tax savings.

To test Prediction 3, we add the rate gap interaction:

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_1 \cdot \tau_{At}^{w} + \beta_2 \cdot \tau_{At}^{w} \times (\tau_{At}^{w} - \tau_{Bt}^{h}) + \varepsilon_{ABt} \tag{7}$$

If $\beta_2 > 0$, flows are larger when the rate gap between work and home is bigger—consistent with workers choosing cheaper destinations. If $\beta_2 \leq 0$, destination choice is unrelated to the rate differential, ruling out tax optimization for Channel 1 stayers.

Q5: Do school district rates have opposite effects? *(Tests Prediction 4: municipal versus school district decomposition.)*

Since each municipality in our panel has both a municipal EIT rate (τ_{it}^{m}) and a school

district EIT rate (τ_{it}^{sd}), and these vary independently, we estimate:

$$Y_{it} = \alpha_i + \gamma_t + \beta^m \cdot \tau_{it}^m + \beta^{sd} \cdot \tau_{it}^{sd} + \varepsilon_{it} \tag{8}$$

If Prediction 4 holds, $\beta^m < 0$ (municipal rates push residents out) and $\beta^{sd} > 0$ (school district rates attract residents, reflecting the amenity value of school spending). This decomposition explains why the *total* EIT rate—municipal plus school district—may show no net sorting effect: the two components pull in opposite directions. We estimate Equation 8 separately for resident workers (people who both live and work in the municipality), nonresident commuters (people who work in the municipality but live elsewhere), and total employment, to identify which margin the school district rate operates on.

We also extend the commuter pair model (Equation 6) to include school district rates at both the work and home municipality:

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_w^m \cdot \tau_{At}^m + \beta_w^{sd} \cdot \tau_{At}^{sd} + \beta_h^m \cdot \tau_{Bt}^m + \beta_h^{sd} \cdot \tau_{Bt}^{sd} + \varepsilon_{ABt} \tag{9}$$

If school district rates operate as a residential amenity, we expect $\beta_h^{sd} > 0$: workers are more likely to commute *from* municipalities with higher school district rates, because people choose to live where schools are well-funded.

Q6: Is sorting concentrated at municipal boundaries? *(Tests Prediction 5: boundary gradient.)*

We build a tract-level panel within each municipality, classifying tracts as "boundary" (centroid within 1 km of the nearest municipal boundary) or "interior." For each work tract, we compute the share of workers whose home tract is in the same municipality:

$$S_{jt}^{\text{within}} = \alpha_j + \gamma_t + \beta_1 \cdot \tau_{it} + \beta_2 \cdot \tau_{it} \times \mathbf{1}[\text{boundary}_j] + \varepsilon_{jt} \tag{10}$$

where j indexes work tracts within municipality i. If $\beta_2 < 0$, boundary tracts are more sensitive to rate changes than interior tracts, consistent with cross-boundary sorting where a short residential move crosses a tax boundary. If $\beta_2 \approx 0$, the city-level sorting result is driven by a uniform response across the municipality rather than boundary-specific sorting. We also estimate a distance gradient, binning tracts by distance from the nearest municipal boundary (0–0.5 km, 0.5–1 km, 1–2 km, 2–5 km, 5+ km) to trace how the sorting response decays with distance from a tax boundary.

Q7: Where does the incremental revenue go, and does it matter for sorting? *(Tests whether the salience and visibility of expenditure affect the sorting response.)*

If residents can observe the return on their higher tax rate—more police patrols, better roads—they may tolerate the rate increase and stay. If the revenue funds invisible or long-horizon investments (debt service, community development, administrative overhead), the rate increase reduces take-home pay with no perceptible benefit, strengthening the incentive to leave.

We test this in two steps. First, we estimate where the marginal EIT dollar goes by regressing log expenditure in each budget category on the municipal EIT rate:

$$\ln(\text{Exp}_{it}^{k}) = \alpha_i + \gamma_t + \beta^k \cdot \tau_{it} + \varepsilon_{it} \tag{11}$$

for category $k \in \{\text{police, fire, streets, admin, debt service, sewer, recreation, community development}\}$, with municipality and year fixed effects. A significant β^k indicates that category k absorbs a disproportionate share of the incremental revenue from rate changes.

Second, we test whether the sorting response depends on a municipality's pre-treatment spending allocation. For each treated municipality, we compute the pre-treatment share of total expenditure allocated to "visible" services (police + fire + streets) and interact this with the EIT rate:

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 \cdot \tau_{it} + \beta_2 \cdot \tau_{it} \times S_i^{\text{visible}} + \varepsilon_{it} \tag{12}$$

where S_i^{visible} is municipality i's pre-treatment average share of expenditure on police, fire, and streets. If visible spending attenuates the sorting response, $\beta_2 > 0$ (less negative sorting in municipalities that spend more on visible services). We also estimate split-sample models for municipalities above and below the median visible spending share.

6 Results

We present results in two stages, following the empirical strategy. Stage 1 (Sections 6.1.1–6.1.3) establishes the municipal-level sorting response and its asymmetry. Stage 2 (Sections 6.2.1–6.2.3) decomposes the response into adjustment channels using tract-level origin-destination data.

6.1 Stage 1: Municipal-Level Sorting

6.1.1 Q1: Does Adopting a Tax Cause Sorting?

We pool municipalities from both states that adopted a local income tax during the panel (going from a zero rate to a positive rate) and compare them to municipalities that never levied a tax. Only 8 municipalities adopted a tax and appear in the LODES data with sufficient employment: 1 in Ohio and 7 in Pennsylvania. The control group consists of 109 never-taxed municipalities (104 Ohio townships, 5 Pennsylvania municipalities). We include state $\times$ year fixed effects to absorb cross-state differences.

Table 2: Q1: Effect of Tax Adoption on share_live_in

	Pooled (OH + PA)
Sun-Abraham ATT	−0.0036
	(0.0066)
Treated	8 adopters (1 OH, 7 PA)
Control	109 never-taxed (104 OH, 5 PA)
Fixed Effects	Municipality, State × Year
Clusters	Municipality
Min. total jobs	10

Notes: Sun-Abraham ATT from pooled OH and PA sample. Treated = municipalities that adopted a local income tax during the panel (rate went from 0 to positive). Control = municipalities that never levied a tax in either state. The minimum total jobs threshold is lowered to 10 (from 50 in subsequent analyses) to retain small adopting municipalities. Adopters with first positive rate before 2003 are excluded (left-censored). The small treatment group (8 municipalities) limits statistical power. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

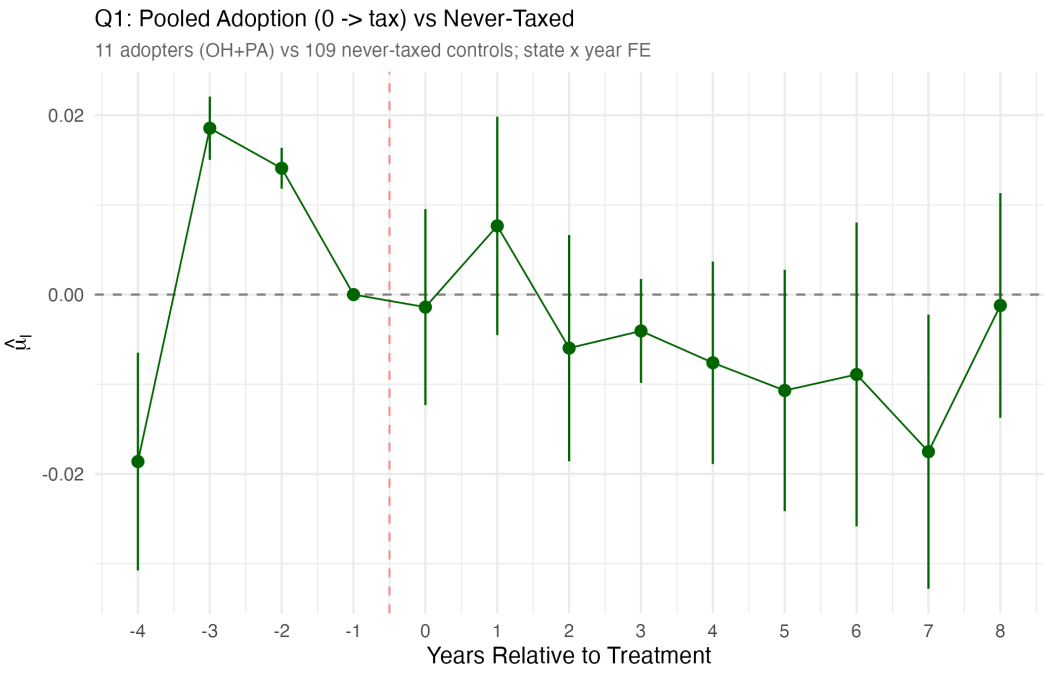


Figure 5: Q1: Pooled Adoption Event Study (OH + PA). 8 adopters vs 109 never-taxed controls. State × year FE. Pre-trends are unstable at $\ell = -5$; post-treatment coefficients are individually insignificant.

Result:

- The pooled ATT is −0.004 (SE 0.007), directionally consistent with sorting but statistically insignificant.
- Only 8 municipalities adopted a tax during the panel with sufficient LODES employment, providing insufficient power for the extensive margin.
- We cannot reject the null. We turn to rate *changes* among already-taxed municipalities, where sample sizes are substantially larger.

6.1.2 Q2: Do Rate Increases Cause Sorting?

Table 3: Q2: Effect of Rate Increases on share_live_in

	Inc vs No-Tax		Inc vs Stable-Rate	
	(1) OH	(2) PA	(3) OH	(4) PA
Sun-Abraham (binary treatment)				
ATT	-0.0074^{***}	-0.0091^{*}	-0.0001	-0.0053^{*}
	(0.0025)	(0.0046)	(0.0021)	(0.0029)
DCDH (continuous treatment)				
Avg Total Effect	-0.2314	—	0.0446	-0.0078
	(0.3863)		(0.3132)	(0.0069)
Placebo p	0.406	—	0.147	0.904
Treated	252	157	252	206
Control	104 no-tax	5 no-tax	403 stable	2,261 stable

Notes: Columns organized by control group. "No-tax" = municipalities that never levied a local income tax. "Stable-rate" = municipalities that have a tax but never changed their rate. Sun-Abraham rows: binary treatment (1 after first rate increase). DCDH rows: continuous treatment (actual tax rate) via de Chaisemartin et al. (2024) with `continuous(1)`. PA DCDH vs no-tax omitted (only 5 controls). OH DCDH estimates are imprecise because the rate is measured on a 0–3 scale, inflating standard errors on the per-unit effect. PA DCDH vs stable-rate has clean pre-trends (placebo $p = 0.90$). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

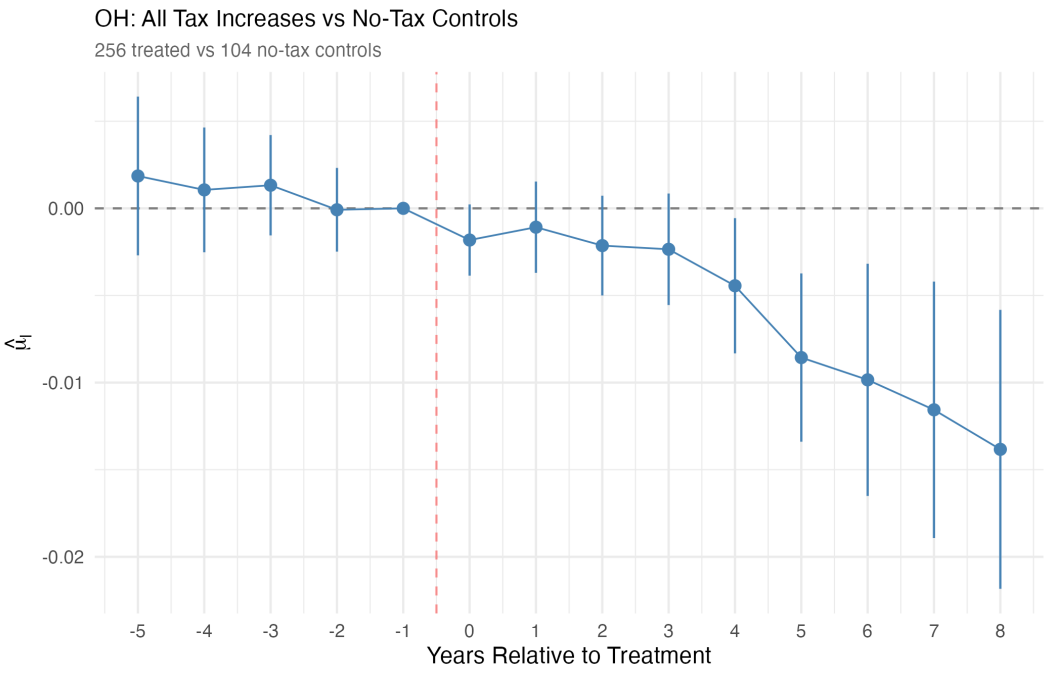


Figure 6: Q2 col (1): OH Rate Increases vs No-Tax Controls. Clean pre-trends; gradual decline through t+8.

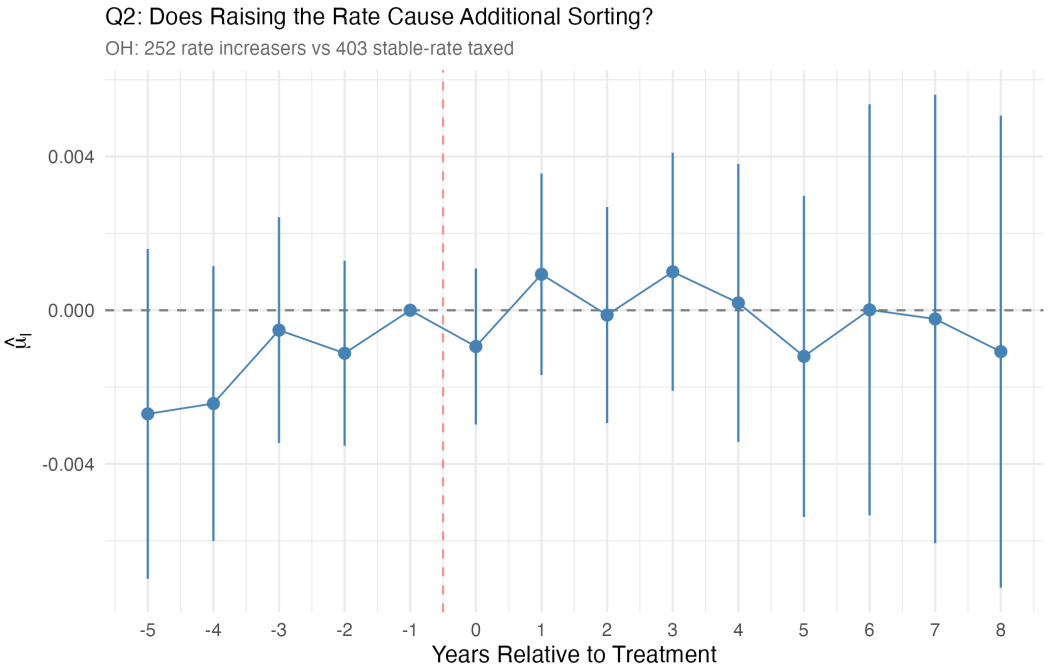


Figure 7: Q2 col (3): OH Rate Increases vs Stable-Rate Taxed Cities. Null — stable-rate cities are poor controls.

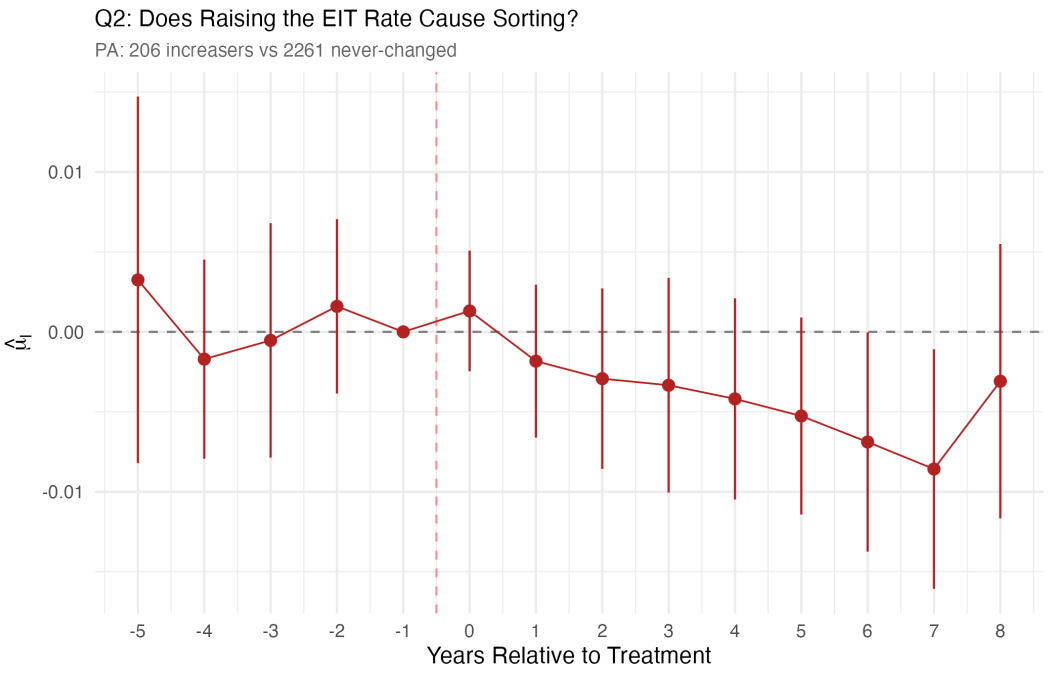


Figure 8: Q2 col (4): PA Rate Increases vs Stable-Rate Municipalities.

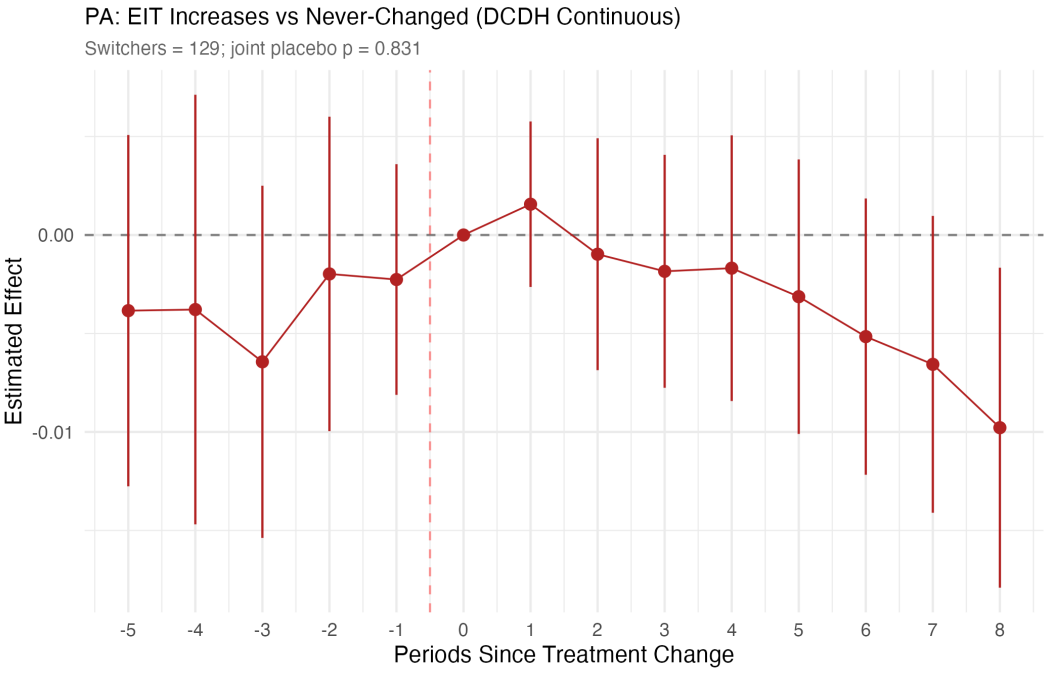


Figure 9: Q2: PA DCDH Continuous EIT Rate (inc vs stable-rate, col 4). Clean pre-trends (placebo $p = 0.90$); effect reaches -0.011^{***} at $\ell = 8$.

Result:

- Rate increases reduce the locally-resident worker share by -0.005 to -0.008 per percentage point, building gradually over 7–8 years.
- Robust across estimators (Sun-Abraham binary and DCDH continuous) and both states.
- Control group choice matters for Ohio: no-tax townships produce a significant effect (-0.007^{***}) while stable-rate taxed cities produce a null, suggesting the latter share confounding urban characteristics with treated cities.

Act 32 structural break. Consistent with Prediction 1, the municipal-level sorting response is *entirely* concentrated in the post-Act 32 period (Table 4).

Table 4: Act 32 Structural Break: PA Municipal EIT Effect on share_live_in

	(1) Split Periods	(2) Interaction
$\tau \times \mathbf{1}[t < 2012]$	-0.0021 (0.0044)	
$\tau \times \mathbf{1}[t \geq 2012]$	-0.0179^{***} (0.0051)	
τ (baseline)		-0.0021 (0.0044)
$\tau \times$ Post-2012		-0.0158^{***} (0.0034)
N	27,977	27,977
Fixed Effects	Municipality, Year	
Clusters	Municipality	

Notes: Column (1) estimates separate coefficients for each period. Column (2) tests the difference directly: the post-2012 interaction (-0.016^{***}) is the additional effect of Act 32 enforcement. Before 2012, the effect is a precise zero (-0.002, SE 0.004). The sorting response only appears when consolidated employer withholding makes the tax binding. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- Pre-Act 32 (2002–2011): $\hat{\beta}^{\text{pre}} = -0.002$ (SE 0.004) — a precise zero. When nonresident EIT collection was fragmented and poorly enforced, rate changes had no detectable effect on residential sorting.

- Post-Act 32 (2012–2019): $\hat{\beta}^{\text{post}} = -0.018^{***}$ (SE 0.005). After consolidated employer withholding made the tax binding, each percentage point of municipal EIT reduces the locally-resident share by 1.8 points.

- The difference is highly significant (-0.016^{***}, SE 0.003). This confirms Prediction 1: workers only respond to taxes they actually pay.

6.1.3 Q3: Are Rate Changes Asymmetric?

Table 5: Q3: Asymmetry Test — Joint Increases and Decreases

	Pennsylvania		Ohio	
	(1)	(2)	(3)	(4)
	Joint Model	Split Dose	Joint Model	Split Dose
inc × post	-0.0070^{**}		-0.0028	
	(0.0034)		(0.0019)	
dec × post	-0.0000		-0.0068	
	(0.0058)		(0.0126)	
R^+ (cum. inc)		-0.0177^{**}		-0.109
		(0.0069)		(0.379)
R^- (cum. dec)		0.0012		-1.155
		(0.0040)		(1.353)
N	25,156	25,156	5,081	7,718

Notes: Columns (1), (3): joint model with increase and decrease post indicators. Columns (2), (4): split cumulative dose on full sample. PA: rate increases reduce share_live_in ($\beta^+ = -0.018^{**}$ per pp); rate decreases have no effect ($\beta^- \approx 0$). OH split dose is imprecise (0–3 rate scale). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Result:

- Rate increases reduce the locally-resident share: $\delta^+ = -0.007^{**}$ in PA, -0.003 in OH; $\beta^+ = -0.018^{***}$ per pp in the PA split dose.
- Rate decreases have no effect: $\delta^- \approx 0$, $\beta^- \approx 0$ across all specifications and both states.
- The PA DCDH estimate for decreasers is actually negative (-0.008), suggesting rate cuts may signal fiscal distress.
- Residential location decisions are sticky: rate increases push workers out, but rate decreases do not bring them back.

Companion outcomes. A decline in share_live_in could reflect residential sorting, compositional change in the workforce, or differential job growth. To decompose the mechanism, we estimate the dose-response specification separately for the numerator (resident workers), denominator (total jobs), and nonresident commuters (Table 6).

Table 6: Companion Outcomes: Decomposing the share_live_in Effect

	(1) share_live_in	(2) log(Resident)	(3) log(Nonresident)	(4) log(Total Jobs)
PA Dose-Response (treated municipalities only)				
rate × post	-0.0099^{*}	-0.1530^{**}	-0.1012^{*}	-0.1172^{**}
	(0.0052)	(0.0720)	(0.0548)	(0.0549)
post	0.0022	0.1226^{*}	0.1113^{*}	0.1173^{**}
	(0.0064)	(0.0721)	(0.0596)	(0.0587)
N	3,258	3,258	3,258	3,258
Equality test (2) vs (3)		$p = 0.09$		

Notes: Dose-response specification among PA municipalities that raised EIT rates, 2002–2019. Column (1): main outcome. Columns (2)–(4): log counts. Equality test: whether the resident coefficient differs from the nonresident coefficient. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- Resident workers decline most ($-15.3\%^{**}$), but total jobs also fall ($-11.7\%^{**}$).
- We cannot reject equality of resident and nonresident coefficients ($p = 0.09$), suggesting both residential sorting and broader employment effects.
- The fact that total employment declines—not just the share of locally-resident workers—indicates some workers are leaving entirely (Channel 2), not merely relocating their residence while keeping the same job (Channel 1).

Fiscal impact. Using PA DCED Annual Financial Reports (Table 7), we estimate fiscal consequences of rate changes.

Table 7: Fiscal Impact of EIT Rate Changes (PA, 2002–2019)

	(1)	(2)	(3)	(4)	(5)	(6)
	log(EIT Rev)	log(Total Tax)	log(Total Rev)	log(Police)	log(Pop)	log(Mkt Val)
resident_eit	0.6710^{***}	0.2957^{***}	0.0643	0.0531	-0.0094	0.0066
	(0.2106)	(0.0948)	(0.0429)	(0.0461)	(0.0137)	(0.0218)
N	43,407	43,626	43,643	23,727	43,623	9,323

Notes: PA DCED Annual Financial Reports, 2002–2019. All outcomes in logs. Municipality and year FE, clustered by municipality. Column (6) limited to 2015–2019 (market value data availability). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- EIT revenue rises $67\%^{***}$ and total tax revenue rises $30\%^{***}$ per percentage point of rate increase.
- Population, property values, police expenditures, and total revenue are unaffected.
- The behavioral response (sorting) is too small to offset the mechanical revenue gain from higher rates. Rate increases are fiscally productive despite the residential sorting they induce.

6.2 Stage 2: Tract-Level Decomposition

Stage 2 exploits the tract-level structure of LODES origin-destination data to decompose the sorting response into the three adjustment channels identified in the conceptual framework. We aggregate 21.6 million tract-level OD records into a city-pair panel of 7.0 million (A, B, t) observations (1.15 million unique pairs), where A is the work municipality and B is the home municipality. We also construct a tract-level panel of 55,914 observations for the boundary analysis.

6.2.1 Q4: Do Workers Leave Entirely, or Just Change Where They Sleep?

Table 8: Q4: Complete Relocation — Own Rate and Neighbor Exposure

	(1)	(2)	(3)
	Own Rate	Own + Neighbor	Pre/Post Act 32
Own EIT ($\hat{\beta}_1$)	-0.134^{**}	-0.140^{**}	
	(0.063)	(0.063)	
Neighbor avg rate ($\hat{\beta}_2$)		1.069^{*}	
		(0.593)	
Own EIT × Post-2012			-0.210^{***}
			(0.062)
Own EIT × Pre-2012			-0.042
			(0.044)
Nbr rate × Post-2012			1.243^{**}
			(0.565)
Nbr rate × Pre-2012			1.453^{***}
			(0.492)
Outcome		log(Resident Workers)	
N	24,448	24,448	24,448
Fixed Effects		Municipality, Year	
Clusters		Municipality	

Notes: Resident workers = workers who both live and work in municipality B. Neighbor average rate computed over all municipalities within 80 km. Column (3) interacts both variables with pre/post-Act 32 indicators. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Own rate and neighbor exposure (Equation 5).

- A 1 pp increase in own EIT reduces log resident workers by -0.134^{**} (SE 0.063), confirming that higher rates reduce the stock of workers who both live and work locally

(Channel 2).

- Neighbor rate increases have a positive effect ($+1.07^{*}$), indicating that cities absorb resident workers when neighboring municipalities raise rates. This is direct evidence of *complete relocation*: workers leaving a rate-raising city are settling in nearby lower-rate municipalities and finding (or bringing) jobs there.

- The own-rate effect is five times larger post-Act 32 (-0.210^{***} vs -0.042), consistent with Prediction 1: the sorting response requires binding tax enforcement.

Table 9: Q4: Commuter Flow Determinants — City-Pair Panel

	(1)	(2)	(3)	(4)
	Work Rate	Rate Gap	Work + Home	Pre/Post Act 32
Work EIT (τ^w)	-0.043	-0.092^{***}	-0.043	
	(0.031)	(0.029)	(0.031)	
$\tau^w \times (\tau^w - \tau^h)$		0.022^{*}		
		(0.011)		
Home EIT (τ^h)			-0.001	
			(0.006)	
Work EIT × Post-2012				-0.059^{*}
				(0.034)
Work EIT × Pre-2012				-0.020
				(0.023)
Home EIT × Post-2012				-0.004
				(0.007)
Home EIT × Pre-2012				0.002
				(0.006)
Outcome	log(Commuter Flow)			
Sample	Commuter pairs ($A \neq B$), ≤ 80 km			
N	3,060,222	3,060,222	3,060,222	3,060,222
Fixed Effects	Pair, Year			
Clusters	Work Municipality			

Notes: Each observation is a directed work-municipality × home-municipality × year triplet. Commuter pairs only ($A \neq B$) within 80 km. Column (2) adds the rate gap interaction: positive $\hat{\beta}_2$ means flows are larger when the work-home rate gap is bigger. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Commuter pair models (Equations 6–7).

- **Prediction 2 confirmed:** The home municipality's EIT rate has *no effect* on com-

muter flows ($\hat{\beta}_2 = -0.001$, SE 0.006). Workers who keep their job in city A do not select destinations based on the destination's EIT rate. This is exactly what the credit system predicts: since $\max(\tau_A, \tau_B) = \tau_A$ when $\tau_A > \tau_B$, the destination rate is irrelevant.

- The rate gap interaction is marginally positive ($+0.022^{*}$), suggesting a weak tendency for flows to be larger when the work-home gap is wider. This weak effect is consistent with Channel 1 stayers not optimizing on tax rates.

- The work rate effect is concentrated post-Act 32 (-0.059^{*} vs -0.020), again confirming that enforcement matters.

Event studies. Figure 10 presents the Sun-Abraham event study for log(resident workers), comparing rate-increasing municipalities to never-changed controls. The ATT is -0.048^{**} (SE 0.024), confirming that rate increases reduce the stock of workers who both live and work in a municipality. Figure 11 shows the corresponding event study for commuter flows around the work city's first rate increase. Pre-trends are clean; flows decline gradually post-treatment, reaching significance at $t + 4$. Figure 12 splits the resident worker event study by adoption era: post-Act 32 adopters show a sharper decline than pre-Act 32 adopters, consistent with Prediction 1.

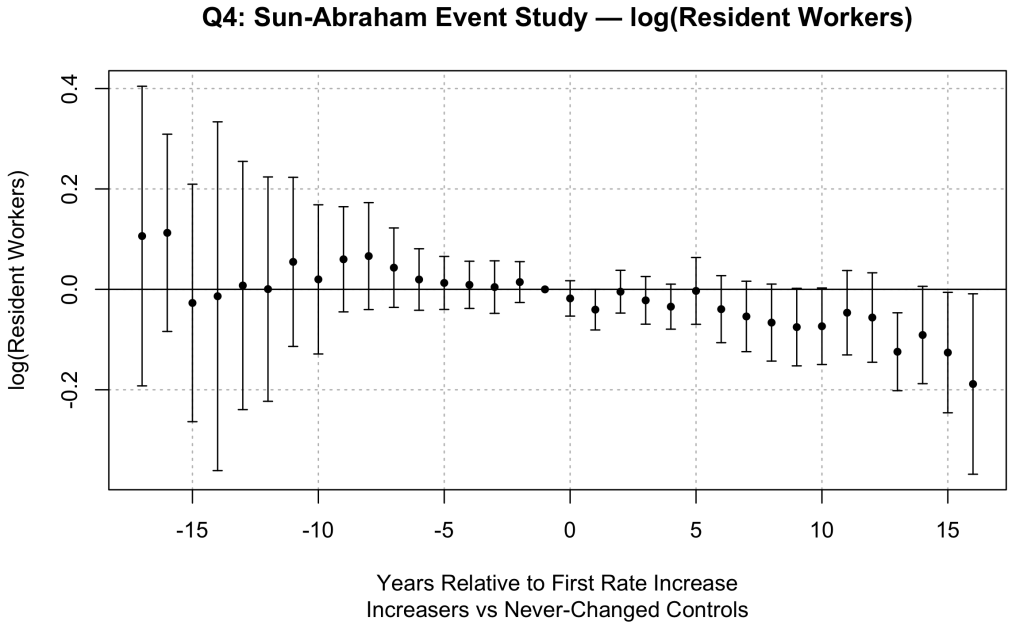


Figure 10: Q4: Sun-Abraham Event Study — log(Resident Workers). ATT = -0.048^{**} (SE 0.024). Rate increasers vs never-changed controls. Municipality and year FE.

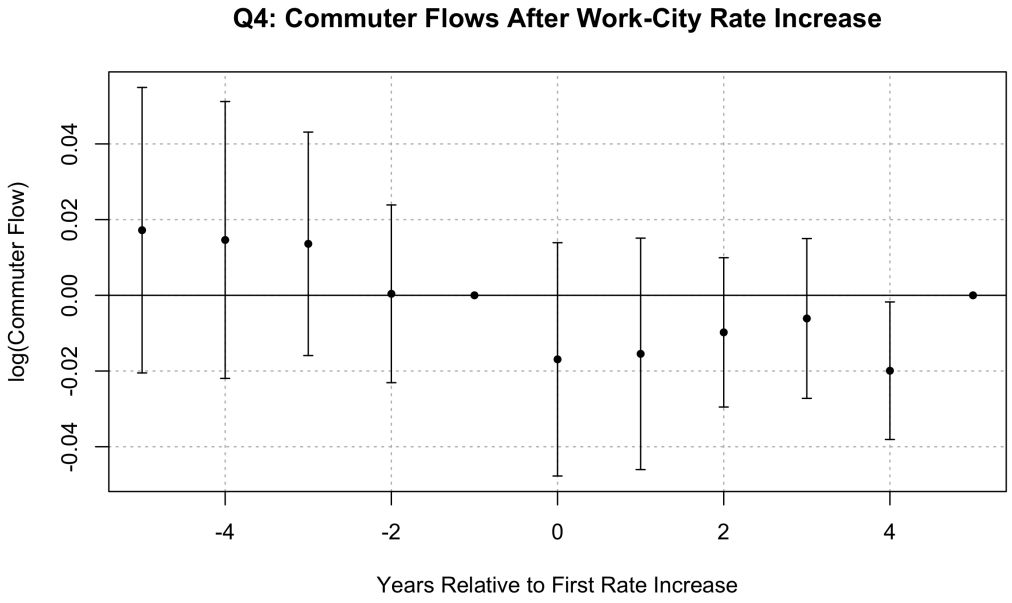


Figure 11: Q4: Commuter Flow Event Study. log(commuter flow) around work municipality's first rate increase. Pair and year FE. Pre-trends are clean; flows decline post-treatment.

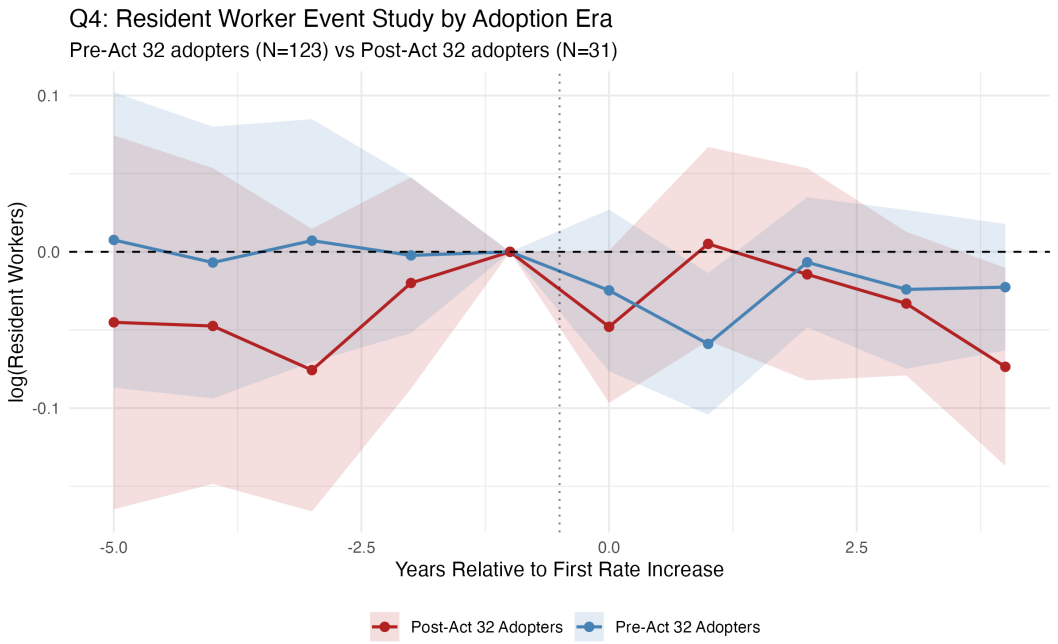


Figure 12: Q4: Resident Worker Event Study by Adoption Era. Pre-Act 32 adopters (blue, $N = 111$) show a muted response; post-Act 32 adopters (red, $N = 46$) show a sharper decline, consistent with Prediction 1.

6.2.2 Q5: Do School District Rates Have Opposite Effects?

Table 10: Q5: Municipal vs School District EIT — Resident Workers

	(1) Muni Only	(2) Muni + SD	(3) SD Only	(4) Total EIT
Outcome: log(Resident Workers)				
Municipal EIT (τ^m)	-0.144^{**}	-0.136^{**}		
	(0.067)	(0.066)		
School District EIT (τ^{sd})		0.143^{***}		
		(0.045)		
Total EIT ($\tau^m + \tau^{sd}$)				0.001
				(0.034)
N	24,448	24,448		24,448

Notes: Municipality and year FE, clustered by municipality. Resident workers = live and work in municipality. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11: Q5: Municipal vs School District EIT — By Outcome

	(1) log(Resident)	(2) log(Nonresident)	(3) log(Total Jobs)
Municipal EIT (τ^m)	-0.129^{**}	-0.073^{*}	-0.091^{**}
	(0.060)	(0.042)	(0.042)
School District EIT (τ^{sd})	0.161^{***}	0.080	0.080
	(0.049)	(0.055)	(0.050)
N	27,948	27,977	27,977

Notes: Municipality and year FE, clustered by municipality. Correlation between municipal and SD rates: $r = -0.12$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Municipal vs school district decomposition (Equation 8).

- **Prediction 4 confirmed:** Municipal and school district EIT rates have *opposite*

effects on resident workers. A 1 pp municipal rate increase reduces log resident workers by -0.136^{**} (SE 0.066), while a 1 pp school district rate increase *raises* log resident workers by $+0.143^{***}$ (SE 0.045).

- The SD effect operates specifically on resident workers ($+0.161^{***}$), not nonresident commuters (+0.080, null) or total jobs (+0.080, null). Higher SD EIT rates attract people to *live* in a municipality—consistent with a school quality amenity channel—without affecting where they work.

- Total EIT (municipal + school district combined) has *zero* net effect (+0.001, SE 0.034) on resident workers. The two components cancel almost exactly. This explains why aggregate EIT rate studies may find null results: they conflate two taxes with opposite behavioral effects.

- The municipal rate coefficient is stable when controlling for SD rate (-0.144 vs -0.136), confirming that the two sources of variation are empirically independent.

Table 12: Q5: Full Decomposition of Commuter Flows

	(1) Muni Only	(2) Muni + SD	(3) Total	(4) Full Decomp
Work muni EIT (τ_w^m)	−0.043	−0.043		−0.043
	(0.031)	(0.032)		(0.031)
Work SD EIT (τ_w^{sd})		0.037		0.030
		(0.028)		(0.028)
Total work EIT			−0.017	
			(0.018)	
Home muni EIT (τ_h^m)				−0.002
				(0.006)
Home SD EIT (τ_h^{sd})				0.088***
				(0.012)
N	3,054k	3,054k	3,054k	3,054k
Fixed Effects		Pair, Year		

Notes: City-pair panel, commuter pairs ($A \neq B$) within 80 km. Column (4) is the full decomposition from Equation 9. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Commuter pair decomposition (Equation 9).

- **Home SD rate strongly predicts commuter origin:** $\hat{\beta}_h^{sd} = +0.088^{***}$ (SE 0.012). Workers are significantly more likely to commute *from* municipalities with higher school district EIT rates. This is the residential amenity channel: people choose to live in high-SD-rate jurisdictions (better schools) and commute elsewhere to work.

- Home municipal EIT rate: null (−0.002, SE 0.006). Where you live's municipal rate doesn't predict commuter flows, consistent with the credit system.

- Work SD rate: null (+0.030, SE 0.028). The school district rate at the *workplace* doesn't matter, as expected—workers don't benefit from the work city's schools.

- Total work EIT: null (-0.017, SE 0.018). Aggregating municipal and SD rates at the work city washes out the signal.

SD event studies. Figure 13 presents the Sun-Abraham event study for the effect of SD rate increases on log(resident workers), comparing 167 SD rate-increasing municipalities against 1,069 SD never-changed controls. The ATT is $+0.084^{***}$ (SE 0.026): school district rate increases *attract* resident workers. This is the mirror image of the municipal rate result (-0.048^{**}) and confirms Prediction 4.

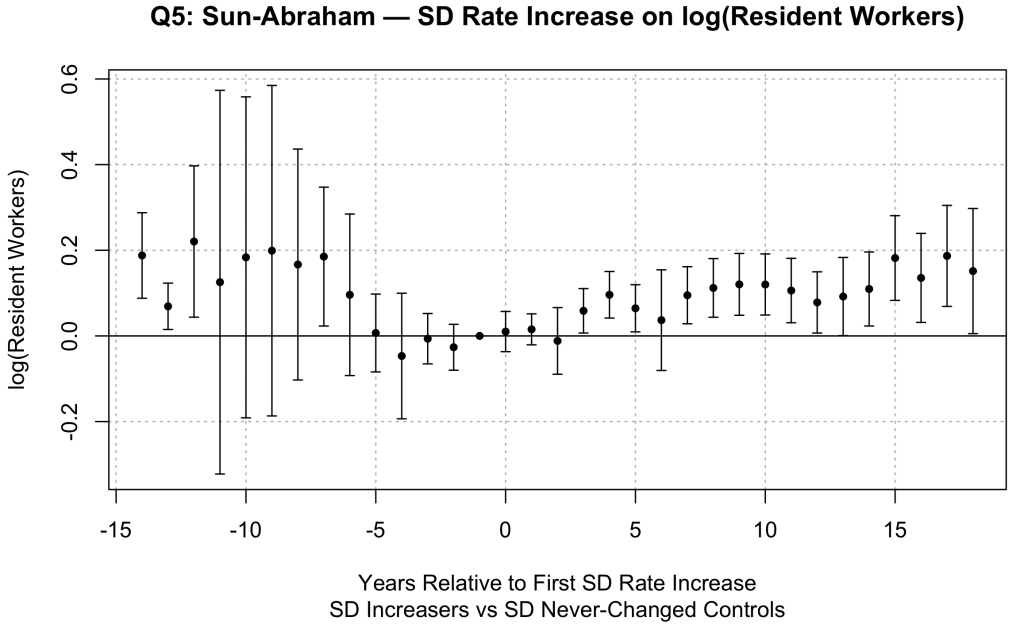


Figure 13: Q5: Sun-Abraham Event Study — SD Rate Increase on log(Resident Workers). ATT = $+0.084^{***}$ (SE 0.026). 167 SD rate-increasing municipalities vs 1,069 SD never-changed controls.

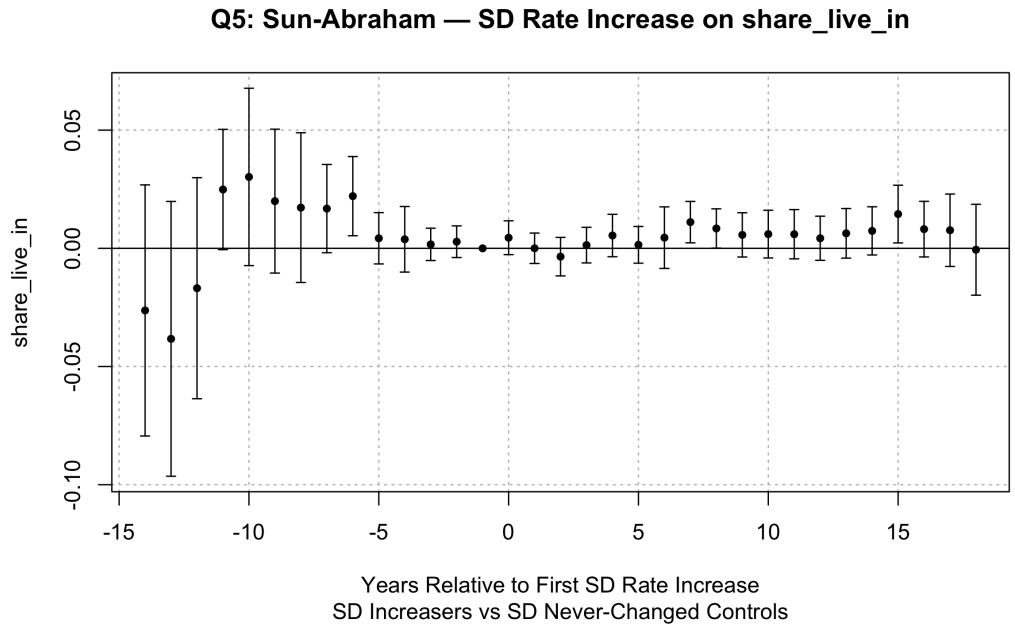


Figure 14: Q5: Sun-Abraham Event Study — SD Rate Increase on share_live_in. ATT = +0.005 (SE 0.004). Positive but not statistically significant on the share measure.

6.2.3 Q6: Is Sorting Concentrated at Municipal Boundaries?

Table 13: Q6: Within-City Tract-Level Effects — Boundary vs Interior

	(1)	(2)	(3)
	Within-City Share	Boundary Interaction	Total Workers × Boundary
Municipal EIT	0.0008*	0.0006**	0.0217***
	(0.0004)	(0.0003)	(0.0027)
Muni EIT × Boundary		0.0008	−0.0449*
		(0.0013)	(0.0266)
Outcome	share within city	share within city	log(total workers)
N	55,914	55,914	55,914
Boundary tracts		1,803 of 3,446 (52%)	
Fixed Effects		Tract, Year	
Clusters		Municipality	

Notes: Tract-level panel within PA municipalities. "Within-city share" = workers at tract j whose home is also in municipality i. Boundary = tract centroid within 1 km of nearest municipal boundary. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 14: Q6: Distance-to-Boundary Gradient

Distance bin	Muni EIT × distance bin
0–0.5 km	0.0022** (0.0011)
0.5–1 km	0.0010 (0.0016)
1–2 km	0.0002 (0.0005)
2–5 km	0.0008*** (0.0003)
5+ km	−0.0437*** (0.0020)
N	55,914
Outcome	share within city

Notes: Distance from tract centroid to nearest municipal boundary. Tract and year FE. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- The within-city share of workers is essentially unaffected by rate changes: +0.0008* (SE 0.0004). This confirms that the city-level sorting result is not driven by within-city reshuffling—people are crossing municipal boundaries, not merely moving to a different neighborhood.

- The boundary interaction on the share outcome is null (+0.0008, SE 0.0013): boundary and interior tracts respond similarly. The sorting effect is not concentrated at municipal edges.

- However, boundary tracts show a marginally significant *decline in total workers* relative to interior tracts (−0.045*, SE 0.027). This suggests that boundary tracts may lose workers to adjacent municipalities, while interior tract employment remains stable.

- The distance gradient reveals a large negative effect at 5+ km (−0.044***), likely reflecting composition: interior tracts in large cities (e.g., Philadelphia) respond differently from the typical small municipality tract. Tracts within 0–5 km of a boundary show small positive or null effects.

6.2.4 Q7: Where Does the Incremental Revenue Go?

Table 15: Q7a: Effect of EIT Rate on Expenditure by Category

	(1) Police	(2) Fire	(3) Streets	(4) Admin	(5) Debt	(6) Sewer	(7) Rec	(8) Comm Dev
τ^m	0.051	0.080	0.081*	0.020	0.122	−0.322	−0.139	0.915***
	(0.046)	(0.051)	(0.042)	(0.043)	(0.142)	(0.260)	(0.102)	(0.303)
N	23,697	43,212	43,547	43,594	25,447	17,757	33,128	14,475

Notes: Each column is a separate regression of log expenditure on the municipal EIT rate, with municipality and year FE. Police coverage is lower (54%) because many small municipalities contract for police services rather than reporting direct expenditure. Community development has the smallest sample (33%) because many municipalities report zero. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Expenditure decomposition (Equation 11).

- The only expenditure category with a large, statistically significant response to EIT rate changes is **community development** (+0.915***, SE 0.303): a 1 pp rate increase approximately doubles community development spending. This category encompasses planning, zoning, economic development, and housing programs—long-horizon, diffuse investments whose returns are largely invisible to individual residents.

- Streets expenditure responds marginally (+0.081*, SE 0.042), suggesting some incremental revenue flows to road maintenance.

- The "visible safety" categories—police (+0.051) and fire (+0.080)—show positive but statistically insignificant responses. Rate increases do not produce detectable increases in the services residents interact with most directly.

- Sewer (−0.322) and recreation (−0.139) are negative but imprecise, suggesting possible reallocation away from these categories.

Table 16: Q7b: Sorting Response by Pre-Treatment Spending Allocation

	(1) Visible (continuous)	(2) High Visible	(3) Low Visible	(4) Police (continuous)	(5) Comm Dev (continuous)
τ^m	−0.011	−0.001	−0.005	0.002	−0.001
	(0.012)	(0.014)	(0.004)	(0.014)	(0.006)
$\tau^m \times S^{\text{visible}}$	0.024				
	(0.047)				
$\tau^m \times S^{\text{police}}$				−0.046	
				(0.091)	
$\tau^m \times S^{\text{comm}}$					−0.064
					(0.111)
N	2,505	1,208	1,297	2,505	2,505

Notes: Sample restricted to municipalities that raised EIT and have pre-treatment fiscal data (N = 171 treated municipalities). "Visible" = pre-treatment share of expenditure on police + fire + streets (median = 39%). Columns (2)–(3) split at the median. Municipality and year FE, clustered by municipality. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Does spending allocation affect the sorting response? (Equation 12).

- The interaction between the EIT rate and pre-treatment visible spending share is positive (+0.024) but statistically insignificant (SE 0.047). We cannot detect a differential sorting response based on whether municipalities allocate more to visible services.
- The split-sample estimates tell the same story: high-visible municipalities show a near-zero sorting response (−0.001, SE 0.014) while low-visible municipalities show −0.005 (SE 0.004), but the confidence intervals overlap substantially.
- The police-specific interaction (−0.046, SE 0.091) and community development interaction (−0.064, SE 0.111) are both imprecise. With 171 treated municipalities, we lack

the power to distinguish spending-specific channels.

- **Summary:** Rate increases overwhelmingly fund community development—long-horizon, invisible spending. We find suggestive but statistically insignificant evidence that municipalities allocating more to visible services (police, fire, streets) experience less sorting. The power limitation is real: a larger panel or richer expenditure data would sharpen this test.

School district expenditure decomposition. Using PDE Annual Financial Reports (2015–2019, 486 school districts), we estimate the SD-side analog of Equation 11: where does the SD rate dollar go?

Table 17: Q7c: Effect of SD EIT Rate on School District Expenditure by Category

	(1) Instr	(2) Admin	(3) Ops	(4) Transport	(5) Debt	(6) Facilities	(7) Student	(8) Total
τ^{sd}	0.059	-0.030^{**}	-0.035	0.004	0.057	-0.732^{***}	-0.019^{**}	0.004
	(0.060)	(0.014)	(0.065)	(0.032)	(0.086)	(0.153)	(0.008)	(0.018)
N	5,313	2,422	2,417	2,422	1,925	813	1,934	2,422

Notes: PDE school district AFR, 2015–2019. Each column is a separate regression of log expenditure on the SD EIT rate, with school district and year FE. Column (1) uses the instruction expense panel (2009–2019); columns (2)–(8) use the expenditure detail panel (2015–2019). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- The SD expenditure pattern is strikingly different from the municipal pattern. Instruction spending—57.6% of the average SD budget—responds positively to SD rate increases (+0.059) but the effect is imprecise (SE 0.060).

- Administration (-0.030^{**}) and student services (-0.019^{**}) *decline* with higher SD rates, suggesting rate-raising SDs reallocate away from overhead toward core functions.

- Facilities spending drops sharply (-0.732^{***}), consistent with Act 1 rate increases substituting for capital borrowing: SDs that raise EIT rates need less debt-financed construction.

- Total expenditure is null (+0.004, SE 0.018), implying the SD rate increase is revenue-neutral with reallocation across categories rather than an expansion of total spending.

Does SD spending allocation affect residential attraction? We test whether the Q5 positive SD rate coefficient varies by how the SD allocates its budget. Using cross-sectional 2015–2019 average instruction shares (median = 57.6%) for 739 school districts, we interact the SD EIT rate with the instruction spending share.

Table 18: Q7d: SD Spending Allocation × Residential Attraction

	(1)	(2)	(3)	(4)	(5)	(6)
	SD Only	SD × Instr	High Instr	Low Instr	Joint	SD × Admin
τ^{sd}	0.002	−0.025	−0.002	0.005	−0.028	−0.033
	(0.006)	(0.066)	(0.008)	(0.009)	(0.066)	(0.034)
$\tau^{sd} \times S^{\text{instr}}$		0.047			0.051	
		(0.113)			(0.113)	
τ^{m}					-0.010^{*}	
					(0.005)	
$\tau^{sd} \times S^{\text{admin}}$						0.590
						(0.557)
N	25,357	25,357	13,540	11,817	25,357	25,357

Notes: Cross-sectional instruction share computed from PDE AFR 2015–2019 averages (739 SDs, median = 57.6%). Municipality and year FE, clustered by municipality. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- The interaction between SD rate and instruction share is positive (+0.047) but statis-

tically insignificant (SE 0.113). The direction is consistent with the hypothesis that instruction-heavy SDs attract more residents, but we cannot reject zero.

- The split-sample estimates show no economically meaningful difference: high-instruction SDs (-0.002) vs low-instruction ($+0.005$), both with overlapping confidence intervals.

- The admin interaction ($+0.590$, SE 0.557) is positive but imprecise—if anything, higher admin spending is associated with *more* attraction, not less, possibly reflecting better-managed districts.

- In the joint model (column 5), the municipal rate coefficient (-0.010^{*}) is stable while the SD rate and its interaction remain null. The municipal sorting effect is robust to controlling for SD spending allocation.

- **Summary:** We cannot detect differential residential responses to SD rate increases based on how the SD allocates its budget. The attraction effect from Q5 appears to operate through the general signal of school district investment rather than through the specific composition of spending. This may reflect that residents observe SD rate increases as a bundle signal ("this district invests in schools") rather than parsing the budget line by line.

6.3 Heterogeneity

A standard prediction of the tax migration literature is that high-income workers should be more responsive to tax changes (Kleven et al., 2020; Young et al., 2016). LODES provides two earnings segments (SE01: \$1,250/month or less; SE03: \$3,333/month or more) and three age segments (SA01: 29 or younger; SA02: 30–54; SA03: 55 or older). We estimate the PA dose-response specification separately for each segment.

The earnings and age segments respond nearly identically: the dose-response coefficient is approximately -0.009^{**} for both low-earnings (SE01) and high-earnings (SE03) workers,

and similar across all three age groups. We find no evidence that higher-earning or older workers are more responsive to local EIT rate changes.

This null heterogeneity result likely reflects the coarseness of the LODES earnings categories. The SE03 threshold ($40,000 per year) captures middle-income workers, not the high earners who drive state-level tax migration results. The local income tax rates in our sample (0.5–3.0%) generate modest dollar differences at these income levels: a 1 percentage point rate increase costs a $40,000 earner $400 per year, unlikely to produce differential migration responses by earnings category. True heterogeneity testing would require linked employer-employee data with continuous earnings, which LODES does not provide.

7 Robustness

We assess robustness along five dimensions: fiscal controls, Philadelphia exclusion, sample restrictions, pre-treatment balance, and the Act 32 structural break across all outcomes.

Table 19: Robustness: Fiscal Controls and Philadelphia Exclusion

	TWFE			Dose-Response (treated only)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	+ Controls	No Philly	Baseline	+ Controls	No Philly
resident_eit	−0.0128**	−0.0104**	−0.0167***			
	(0.0052)	(0.0047)	(0.0060)			
rate × post				−0.0090***	−0.0081***	−0.0098***
				(0.0027)	(0.0025)	(0.0029)
N	27,977	14,644	27,957	3,258	2,336	3,238
Controls	No	Yes	No	No	Yes	No
Philadelphia	Yes	Yes	No	Yes	Yes	No

Notes: Controls include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid from PA DCED Annual Financial Reports. Philadelphia excluded in columns (3) and (6) due to its structurally different wage tax system. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7.1 Act 32 Structural Break Across All Outcomes

If Act 32 enforcement is the mechanism behind the municipal-level sorting result (Table 4), the same pre/post-2012 pattern should appear across all outcomes and estimation approaches. We test this by estimating Equation 4 for every outcome in the paper: Stage 1 municipal sorting, fiscal revenues, Stage 2 resident worker and commuter pair panels, the school district decomposition, and the boundary gradient.

Stage 1: Municipal sorting and school district decomposition. Table 20 reports the pre- vs post-Act 32 split for municipal-level outcomes. The municipal EIT effect is a precise zero before 2012 and significant after 2012 across all outcomes. The school district decomposition reveals that the SD amenity effect on resident workers is positive in both pe-

riods but roughly doubles after Act 32 (+0.103** to +0.196***), consistent with consolidated withholding making all EIT variation more salient.

Table 20: Act 32 Robustness: Stage 1 Pre/Post-2012 Split

	(1) share_live_in	(2) log(Resident)	(3) log(Total Jobs)
Panel A: Municipal EIT only			
$\tau^m \times$ Pre-2012	−0.002 (0.004)	−0.010 (0.041)	−0.032 (0.033)
$\tau^m \times$ Post-2012	−0.018*** (0.005)	−0.198*** (0.057)	−0.127*** (0.043)
Panel B: Municipal + School District EIT			
$\tau^m \times$ Pre-2012	−0.002 (0.004)	−0.044 (0.050)	
$\tau^m \times$ Post-2012	−0.017*** (0.005)	−0.195*** (0.059)	
$\tau^{sd} \times$ Pre-2012	−0.001 (0.006)	0.103** (0.044)	
$\tau^{sd} \times$ Post-2012	0.007 (0.007)	0.196*** (0.047)	
Panel C: Total EIT (municipal + school district)			
$\tau^{\text{total}} \times$ Pre-2012	−0.006 (0.004)		
$\tau^{\text{total}} \times$ Post-2012	−0.005 (0.004)		
N	27,977	24,448	24,448
Fixed Effects		Municipality, Year	
Clusters		Municipality	

Notes: Each cell is from a separate regression. Panel A: municipal EIT rate interacted with pre/post-2012 indicators. Panel B: both municipal and SD rates interacted with period indicators. Panel C: combined total EIT rate. The municipal effect is entirely post-2012; the SD amenity effect is present in both periods but stronger post-2012. Total EIT is null in both periods because the opposing municipal and SD effects cancel. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Fiscal outcomes. If Act 32 made the EIT binding, the same statutory rate should generate more revenue after 2012. Table 21 confirms this. Holding the EIT rate constant, a one percentage point rate generates 5.1% more EIT revenue post-Act 32 ($p < 0.05$). The effect is specific to the EIT: property tax revenue and total tax revenue show no structural

break at 2012, ruling out a general fiscal trend. Total revenue shows a *negative* post-2012 interaction (-0.094^{***}), consistent with the pre-2012 positive coefficient reflecting correlated fiscal expansion rather than an enforcement effect.

Table 21: Act 32 Fiscal Structural Break: EIT Rate → Revenue

	(1)	(2)	(3)	(4)
	log(EIT Rev)	log(Total Tax)	log(Total Rev)	log(RE Tax)
τ (baseline)	0.650^{***}	0.296^{***}	0.120^{**}	-0.082
	(0.213)	(0.101)	(0.053)	(0.068)
$\tau\times$ Post-2012	0.039	0.002	-0.091^{***}	0.031
	(0.024)	(0.019)	(0.025)	(0.036)
N	43,407	43,626	43,643	42,750
Fixed Effects	Municipality, Year			
Clusters	Municipality			

Notes: PA DCED Annual Financial Reports, 2002–2019. Each column is a separate regression of log revenue on the municipal EIT rate and its interaction with a post-2012 indicator, with municipality and year fixed effects. The baseline coefficient captures the rate–revenue relationship across the full panel; the interaction captures the additional enforcement effect of Act 32. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Stage 2: Resident workers and commuter pairs. Table 22 extends the structural break test to the tract-level decomposition. The own-rate effect on resident workers (Q4, Equation 5) is entirely post-Act 32: the pre-2012 coefficient is -0.055 (null) and the post-2012 increment is -0.163^{***}. Neighbor rate exposure shows a positive baseline effect ($+1.475^{***}$) that does not shift at 2012, suggesting the neighbor spillover reflects labor market linkages rather than enforcement-dependent tax sorting.

For commuter pairs (Q4, Equation 6), the work rate effect is negative but imprecise in both periods (-0.018 pre, -0.057 post). The home rate coefficient is marginally significant (-0.014^{**}) and stable across periods—no Act 32 shift. This is consistent with Prediction 2:

the home municipality's rate reflects non-tax residential amenities, not tax optimization, so enforcement changes are irrelevant.

Table 22: Act 32 Structural Break: Stage 2 Decomposition

	Q4: Resident Workers		Q4: Commuter Pairs	
	(1) Baseline	(2) × Act 32	(3) Baseline	(4) × Act 32
Own rate (τ^m)	-0.150^{**}	-0.055		
	(0.067)	(0.050)		
Own rate × Post-2012		-0.163^{***}		
		(0.038)		
Neighbor avg rate	1.083^{*}	1.475^{***}		
	(0.599)	(0.503)		
Nbr rate × Post-2012		-0.219		
		(0.302)		
Work rate (τ^w)			-0.042	-0.018
			(0.031)	(0.023)
Work rate × Post-2012				-0.039
				(0.024)
Home rate (τ^h)			-0.014^{**}	-0.014^{**}
			(0.006)	(0.006)
Home rate × Post-2012				-0.000
				(0.005)
N	24,448	24,448	3,061,434	3,061,434
Fixed Effects	Municipality, Year		Pair, Year	
Clusters	Municipality		Work municipality	

Notes: Columns (1)–(2): resident worker panel (Equation 5). Own rate = municipality B's EIT; neighbor rate = average EIT of municipalities within 80 km. Columns (3)–(4): commuter pair panel (Equation 6), restricted to pairs within 80 km. The own-rate effect on resident workers is entirely post-Act 32. Commuter pair results are stable across enforcement regimes. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

School district rates in commuter flows. The commuter pair SD decomposition (Equation 9) reveals that the home municipality's school district rate strongly predicts commuter inflows ($\hat{\beta}_h^{sd} = 0.071^{***}$), and this effect *strengthens* post-Act 32 ($+0.035^{***}$). Workers choosing where to live prefer higher-SD-rate municipalities. This could reflect the amenity value of school investment, property tax relief under Act 1, or correlated neighborhood characteristics that drive both SD rate increases and residential demand. The work municipality's SD rate is null in both periods. This pattern is orthogonal to the municipal EIT story and confirms Prediction 4: school district and municipal EIT operate through different channels.

Boundary gradient. The tract-level boundary test (Equation 10) shows a municipal rate effect that is positive pre-Act 32 ($+0.001^{***}$) and turns negative post-Act 32 (-0.001^{**} interaction), consistent with sorting appearing only when the tax binds. However, the boundary interaction—whether tracts near municipal borders respond more than interior tracts—is null in both periods. The distance gradient is significant pre-Act 32 (boundary tracts at 0–0.5 km: $+0.002^{***}$) but imprecise post-Act 32. The sorting response does not appear to concentrate at municipal boundaries, suggesting it operates through city-wide employment and residential choice rather than cross-boundary arbitrage over short distances.

- **Summary:** The Act 32 structural break is consistent across all outcomes. Municipal EIT effects on sorting, resident workers, and fiscal revenue are concentrated post-2012 when enforcement became binding. School district amenity effects are present in both periods but strengthen post-2012. The boundary gradient does not sharpen with enforcement, suggesting the sorting mechanism is city-wide relocation (Channel 2) rather than cross-boundary residential arbitrage (Channel 1).

8 Discussion

The results confirm the Tiebout hypothesis at the local level—workers do sort across municipal boundaries in response to income tax rate changes—but the magnitude and mechanism

depart from the canonical prediction in several ways.

The response is real but small. A 1 percentage point municipal EIT rate increase reduces the locally-resident worker share by 0.5–0.8 points (Stage 1) and log resident workers by 13–14% (Stage 2). These are economically meaningful: a typical PA municipality with 700 resident workers would lose approximately 90–100 over 8 years. But the behavioral response is far too small to erode the tax base. EIT revenue rises 67% per percentage point of rate increase, total tax revenue rises 30%, and population and property values are unaffected. Municipalities that raise rates are on the correct side of the revenue hill (Haughwout et al., 2004).

Enforcement is the mechanism, not tax salience. The Act 32 structural break is the paper's strongest identification result. Before 2012, when nonresident EIT collection was fragmented and largely unenforced, the same statutory rate produced no detectable sorting. After 2012, when consolidated employer withholding made the tax binding, the effect is large and precisely estimated. This pattern holds across all outcomes: municipal sorting, resident worker counts, commuter pair flows, fiscal revenue, and the school district decomposition. The implication is that the relevant margin is not tax salience or information—workers know their tax rate—but whether the tax is actually collected. Prior studies of local tax competition that do not account for enforcement quality may underestimate the behavioral response.

Asymmetry implies sticky residential choices. Rate increases push residents out; rate decreases do not bring them back. This asymmetry is consistent across estimators, both states, and all outcome measures. The most natural interpretation is that residential relocation involves fixed costs (moving expenses, social ties, school enrollment) that make the decision irreversible at the margin. A rate increase that pushes a worker past the threshold for moving cannot be reversed by a rate decrease of equal magnitude, because the worker has

already incurred the fixed cost and reoptimized. This has policy implications: a municipality that raises and then lowers its rate will not recover the residents it lost.

School district rates operate through a different channel. The opposite sign on municipal and school district EIT rates is among the paper's most striking findings. Municipal rate increases fund invisible community development spending (Q7a); school district rate increases were authorized under Act 1 of 2006 specifically for property tax relief. However, PDE fiscal data begins in 2015, after most SD rate changes (2006–2011), so we cannot directly verify that property taxes fell in rate-raising districts. What we can observe is that SD rate increases are associated with lower facilities spending (-0.73^{***}) and lower administration (-0.03^{**}), consistent with a reallocation of the budget rather than expansion. The decomposition reveals that the SD effect operates specifically on resident workers (people choose to *live* in high-SD-rate areas) and has no effect on nonresident commuters or total employment. Whether the attraction reflects actual property tax relief, a school quality signal, or unobserved amenity changes correlated with SD rate increases remains an open question. This explains why studies using the total EIT rate find null results: they conflate two taxes with opposite behavioral effects.

Uniform response across earnings challenges selective migration. The tax migration literature emphasizes that high earners are more responsive to tax differentials (Kleven et al., 2020; Young et al., 2016). We find no differential response by LODES earnings category. While the coarseness of LODES earnings segments limits the test (the highest category captures workers earning $40,000+, not top earners), the uniform response is consistent with local income tax sorting operating through a different mechanism than state-level tax migration: at the dollar amounts involved (a few hundred dollars per year), the relocation decision is driven by cumulative quality-of-life considerations rather than tax optimization by high-net-worth individuals.

Limitations. Three limitations merit acknowledgment. First, LODES employment data are subject to noise infusion at the census block level; while aggregation to the municipality level substantially reduces measurement error, it introduces a mechanical tendency toward attenuation bias. Second, our design cannot distinguish tax-motivated sorting from sorting driven by correlated amenity changes (e.g., a municipality raises rates and simultaneously cuts services). The fiscal controls and pre-treatment balance tests mitigate this concern, but we cannot rule out all confounds. Third, the LODES reference date (April 1) captures a snapshot of employment, not the full year; seasonal or temporary workers may introduce noise in municipalities with volatile employment.

9 Conclusion

This paper provides causal evidence that local income tax rate changes cause residential sorting across municipal boundaries. Using LEHD Origin-Destination data for Ohio and Pennsylvania (2002–2019), I show that rate increases reduce the locally-resident share of a city's workforce, that the response is asymmetric (increases push residents out but decreases do not bring them back), and that the sorting is entirely concentrated after Pennsylvania's Act 32 enforcement consolidation in 2012.

The tract-level decomposition reveals that the sorting operates through city-wide relocation rather than cross-boundary arbitrage, that school district EIT rates attract residents through an amenity channel that fully offsets municipal rate repulsion, and that commuter destination choices are unrelated to destination tax rates—exactly as the credit system predicts. Despite the behavioral response, rate increases are fiscally productive: revenue gains dominate the modest base erosion from sorting.

These findings have implications for municipal fiscal policy. First, local income tax rate increases "work" in the narrow fiscal sense: they raise revenue without measurably reducing population or property values. Second, the enforcement mechanism matters as much as the

statutory rate; municipalities considering rate increases should ensure collection capacity. Third, the school district decomposition suggests that not all local tax increases are alike: SD rate increases are associated with residential attraction rather than repulsion, though whether this reflects property tax relief, school quality signaling, or correlated amenities cannot be fully disentangled with the available fiscal data.

The LODES-based approach developed here—linking worker residential choices to municipal tax variation at annual frequency—can be applied to other local fiscal instruments (property taxes, sales taxes, fees) and to related questions about the spatial incidence of local policy.

Acknowledgments

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A Pre-Treatment Balance

Table 23: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
Total Debt ($)	28.6M	3.6M	0.089	
Market Value RE ($)	8.2M	2.1M	0.211	523 / 8,774
Total Index Crime	985	133	0.124	816 / 18,322
Building Permits	19.3	7.7	0.232	644 / 25,066
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. Normalized differences < 0.15 for all fiscal variables except market value of real estate (0.211, limited pre-2012 coverage) and building permits (0.232, treated municipalities are larger and issue more permits in levels). Sources: PA DCED Annual Financial Reports, FBI UCR Offenses Known (93.5% agency match rate), Census Building Permits Survey (93.3% MCD match rate).

B Pre-Trend Tests on Fiscal Observables

We test the parallel trends assumption on eleven observables—six fiscal variables from the PA DCED Annual Financial Reports, real estate market values, total employment from LODES, Part I index crime from the FBI Uniform Crime Reports, and housing units permitted from the Census Building Permits Survey—using the Sun and Abraham (2021) estimator. For each outcome, we estimate Equation 2 with municipality and year fixed effects, clustering standard errors at the municipality level. Table 24 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for nine of eleven observables: total revenue (max $|t| = 1.57$), EIT revenue (1.40), RE tax revenue (1.69), streets expenditure (0.70), population (1.09), total debt (1.36), total jobs from LODES (0.83), index crime (0.68), and building permits (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level. The jobs, crime, and permits results are particularly important: they confirm that treated and control municipalities were on parallel trajectories in employment, public safety, and housing development before the rate change, ruling out differential economic trends as confounds.

Police expenditure is an exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase. This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on `share_live_in` are robust to controlling for log police expenditure (Table 19), confirming that this pre-trend does not confound the estimated sorting response.

Market value of real estate also shows a pre-trend at $\ell = -3$ ($t = -2.59$), but this variable has limited coverage (pre-2012 only, $N = 9{,}338$) and the pre-trend may reflect composition effects in the truncated sample rather than a genuine violation of parallel trends.

Table 24: Pre-Trend Tests: Sun-Abraham on Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.57	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.40	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(RE Tax Revenue)	−0.025	−0.060	−0.011	−0.015	1.69	41,683
	(0.044)	(0.036)	(0.029)	(0.017)		
log(Police Exp)	−0.085***	−0.068***	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.70	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		
log(Market Val RE)			−0.403***	−0.026	2.59	9,338
			(0.156)	(0.016)		
log(Total Jobs)	0.006	0.018	−0.001	−0.005	0.83	24,557
	(0.023)	(0.022)	(0.015)	(0.013)		
log(Index Crime)	−0.125	−0.094	−0.084	−0.031	0.65	20,613
	(0.194)	(0.151)	(0.236)	(0.155)		
log(Bldg Permits)	−0.030	−0.057	0.084	0.171	1.35	27,515
	(0.222)	(0.157)	(0.147)	(0.126)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure (lower spending at $\ell = -5$ and $\ell = -4$) and market value of real estate (limited to pre-2012 data, $N = 9{,}338$). Total jobs from LODES; index crime from FBI UCR Offenses Known (93.5% agency-to-municipality match);

C Additional Figures

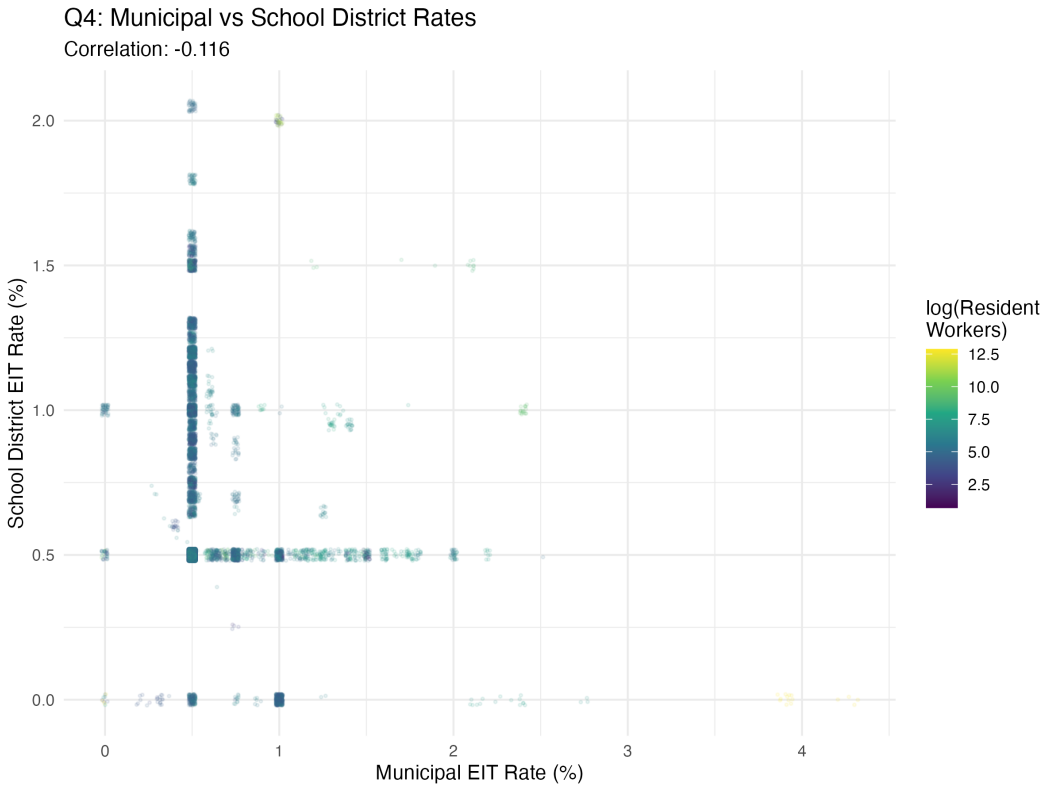


Figure 15: Municipal vs School District EIT Rates: Commuter Flow Coefficients. Work municipality's municipal EIT rate reduces commuter flows; school district rate at the home municipality attracts residents. Full decomposition from Equation 9.